

FEDERAL PUBLIC SERVICE COMMISSION

Aga Khan Road, F-5/1, Islamabad, Pakistan

REQUEST FOR PROPOSAL

Software Development Services for
**Digitalization of In-House Processes and
Automation of Examination System of FPSC**

RFP Reference No: **FPSC/PISC/SW/2026/01**

Issue Date: **20 May 2026**

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Pre-Bid Conference: **3 June 2026, 02:30 PM (PST)**

RFP prepared under Project Implementation and Supervision Consultancy by:
National University of Sciences and Technology (NUST), Islamabad

CLASSIFICATION: CONFIDENTIAL

1. Introduction and Definitions

1.1 Procuring Agency

The Federal Public Service Commission (FPSC) is the premier federal institution established under Article 242 of the Constitution of Pakistan, entrusted with the selection of qualified human resources for the Federal Services through fair, transparent, and open competition. FPSC conducts tests and examinations for recruitment to all public services, the civil services of the Federation, and civil posts in connection with the affairs of the Federation.

FPSC Headquarters is located at Aga Khan Road, F-5/1, Islamabad, with Provincial Offices at Karachi, Lahore, Peshawar, and Quetta, and Regional Offices at Gilgit, Multan, and Sukkur.

1.2 Purpose of this RFP

FPSC invites sealed proposals from state-owned organisations for the design, development, deployment, testing, and maintenance of an integrated software platform for the Digitalization of In-House Processes and Automation of the Examination System of FPSC, as approved under PC-1.

The purpose of this project is to develop Enterprise Resource Planning (ERP) system for the digitalisation of in-house process of FPSC, automation of examination systems, development of question databanks (QDB), and implementation of computer-based testing (CBT).

This RFP covers the software component only. Hardware, network infrastructure, data center hosting, civil works, furniture, and question content creation are procured under separate tenders. The engaged state owned organization will provide the requirement of hardware, software licensing etc. for operating the developed software.

1.3 Applicable Rules

This procurement is governed by the Public Procurement Regulatory Authority (PPRA) Rules 2004 and any subsequent amendments. All provisions of this RFP shall be interpreted in accordance with PPRA Rules. In case of any conflict between this RFP and PPRA Rules, the PPRA Rules shall prevail.

1.4 Definitions

The following terms are used throughout this document:

Term	Definition
FPSC	Federal Public Service Commission, the procuring agency.
PISC	Project Implementation and Supervision Consultant: National University of Sciences and Technology (NUST), engaged by FPSC under a separate contract to provide independent third-party technical validation and oversight.
Vendor / Bidder	The state owned organization submitting a proposal in response to this

Term	Definition
	RFP. After contract award, referred to as the Vendor.
System	The integrated software platform comprising all six modules, website, and candidate portal as specified in Section 3.
PSC	Project Steering Committee: chaired by Secretary/PAO FPSC, responsible for strategic decisions.
PMC	Project Management Cell: operational unit within FPSC IT Wing, responsible for day-to-day project coordination.
PD	Project Director: FPSC officer (BS-20) heading project execution.
EMS	Examination Management System: collective term for Modules 1 through 3 and Module 6, automating FPSC's internal recruitment and examination processes.
CBT	Computer-Based Testing: secure examination delivery system for conducting MCQ-based tests at designated lab locations.
QDBMS	Question Data Bank Management System: secure repository for question lifecycle management including authoring, review, approval, psychometric analysis, and paper generation.
QDB	Question Data Bank: the database containing all question items managed by the QDBMS.
GR	General Recruitment: FPSC's recruitment stream for ex-cadre posts (BPS-16 and above).
CE	Competitive Examination: FPSC's recruitment stream for cadre posts, including the CSS examination.
UEM	Unified Examination Module: configurable module for other examination types conducted by FPSC (FPOE, SOPE, Civil Judges, GB Competitive Exams, etc.).
SEB	Safe Exam Browser: lockdown browser software that restricts candidate access to only the exam interface during CBT.
NTC	National Telecommunication Corporation: government entity providing data center co-location and DR hosting services for this project.
MPLS	Multi-Protocol Label Switching: dedicated secure network circuit connecting CBT sites to the NTC data center.
VPN	Virtual Private Network: encrypted tunnel for secure communication between sites.
HCI	Hyper-Converged Infrastructure: virtualization platform abstracting physical servers into a shared compute, storage, and network pool.
MDM	Mobile Device Management: software controlling and securing endpoint devices (desktops/laptops) at CBT labs.
SAN	Storage Area Network: dedicated high-speed storage system connected to database servers.
DR	Disaster Recovery: secondary site (NTC Lahore) maintaining replicated systems for business continuity.
HA	High Availability: system design ensuring continuous operation through redundancy and failover.

Term	Definition
RPO	Recovery Point Objective: maximum acceptable data loss measured in time (e.g., RPO of 15 minutes means at most 15 minutes of data may be lost in a disaster).
RTO	Recovery Time Objective: maximum acceptable time to restore system operation after a disaster.
NADRA	National Database and Registration Authority: provides biometric identity verification services.
PPRA	Public Procurement Regulatory Authority: regulatory body governing public procurement in Pakistan.
BPR	Business Process Re-engineering: systematic redesign of FPSC workflows to be performed by the Vendor before software development, as required by the PC-1.
SRS	Software Requirements Specification: detailed functional and non-functional requirements document to be produced by the Vendor after BPR.
UAT	User Acceptance Testing: formal testing by FPSC end-users to validate the system meets requirements before go-live.
OWASP	Open Web Application Security Project: international standard for web application security (Top 10 vulnerabilities list).
ISO 27001	International standard for Information Security Management Systems (ISMS).
SSDLC	Secure Software Development Lifecycle: development methodology incorporating security practices at every stage.

1.5 Document Structure

This RFP is organized as follows:

Section	Description
Section 1	Introduction, definitions, and document guide (this section).
Section 2	Project background, current state, objectives, and stakeholder roles.
Section 3	Scope of work: detailed functional requirements for all six modules, website, and candidate portal.
Section 4	Deployment architecture: 4-tier hybrid model, server allocation, site infrastructure, and exam-day operations.
Section 5	Non-functional requirements: performance, availability, security, scalability, backup, data migration, accessibility.
Section 6	Technical requirements: technology stack, source code, IP, development standards, software architecture.
Section 7	Project milestones, deliverables, phasing, training, and acceptance criteria.
Section 8	Vendor eligibility: mandatory pass/fail qualification criteria.
Section 9	Technical evaluation criteria: competitive scoring matrix.
Section 10	Proposal submission requirements: two-envelope method, contents, deadlines.

Section	Description
Section 11	Evaluation methodology: QCBS formula, weightages, award process.
Section 12	Contractual terms: duration, payments, IP, warranty, confidentiality, termination.
Section 13	Annexures: compliance matrix template, financial proposal template, CV template, and draft contract template.

2. Project Background

2.1 Current State

FPSC conducts multiple types of examinations including CSS Competitive Examination, GB Competitive Examination, General Recruitment (GR), Final Passing Out Examination (FPOE), Section Officers Promotion Exam (SOPE), Section Officers APT Exam, Gilgit-Baltistan Recruitment Exam, PMS-to-PAS Induction Exam, Regularization Exam, and any other examination referred by the Federal Govt. Approximately 0.8 million applications are received per year. CSS is conducted annually; GR is an ongoing activity throughout the year.

The recruitment cycle is predominantly manual. A typical cycle takes 150 to 450 days for GR examinations. The 12-step process spans requisition receipt through final nomination. All pre-exam processes are automated except requisition management. The remaining steps (requisition management, advertisement preparation, question paper generation, conduct of examination, result compilation, eligibility scrutiny, interview scheduling, and final nomination, appeal processing and restoration etc.) rely on manual workflows, physical file movement, and paper-based processes.

The processes generating the most delays are:

- Question paper preparation and result compilation, which are dependent on external examiners, manual fictitious numbering, and physical handling of answer books.
- Eligibility scrutiny, which involves manual document verification of every shortlisted candidate.
- Inter-departmental approvals, which rely on physical file movement between wings and directorates.

The following table maps each identified delay in the current recruitment cycle to its root cause, the system capability that addresses it, and the corresponding requirement reference.

Delay Identified	Root Cause	Proposed Solution	Document Traceability Reference
Question paper preparation delays	Manual examiner coordination, physical paper handling, fictitious numbering	QDBMS, automated paper generation, encrypted CBT delivery	Section 3.6 – “CBT & Question Data Bank Management System (QDBMS)” – secure digital paper management and encrypted CBT delivery; Section 4.6 – “Recruitment & Examination Operations Cycle” – automated and secure examination workflows
Result compilation	Manual OMR	Automated CBT	GR-1.6 (OMR scanning

delays	handling, physical answer books, and manual tabulation	scoring and result compilation	and result compilation); CE-2.4 (MPT result preparation); Module 5B (CBT automated scoring and result compilation).
Eligibility scrutiny delays	Manual document verification and candidate scrutiny	Automated scrutiny and rule-based verification	GR-1.7 (automated document scrutiny); CE-2.8 (NADRA Verisys verification); CC-03 (rule-based validation services).
Inter-departmental approval delays	Physical file movement and disconnected workflows	eCase workflow automation and digital approvals	GR-1.12 – “Approval Workflow Management”; CE-2.10 – “Interactive eCase Flow Management”; TAB-04 – “Digital Case Tracking”
Candidate communication delays	Manual notifications and correspondence	Centralized notification engine	CC-02 – “Notification & Communication Services”; MOB-04 – “Mobile Alerts & Notifications”
Lack of centralised records	Excel/Word-based distributed processing	Unified ERP and centralised EMS database	Section 3 – “Enterprise Management System (EMS) Modules”; Section 4 – “System Architecture & Centralised Platform”
Weak operational visibility	No centralised monitoring/dashboard	Executive dashboards and analytics	SUP-6.6 – “Decision Support System (DSS)”; TAB-01 to TAB-05 – “Executive Dashboard Services”
Weak auditability	Manual processing and fragmented approvals	Audit trail and workflow logging	CC-01 – “Audit Trail & Workflow Logging”

Table: Delay-to-Solution Traceability Matrix

2.2 Existing Systems and Data

The Vendor shall account for the following when planning data migration and system design:

- FPSC operates a customised Online Recruitment System with Oracle 11g and MySQL databases at the backend. This system handles online applications, admission certificates, roll number assignment, marksheets, and centre-change processing.
- Cumulative database size is approximately 100 GB, with 17 years of historical application data. Candidate photographs are retained for 2 years only. The database grows at approximately 8 GB per year.
- Post-examination processing at each wing/directorate is maintained separately, mostly in Excel or Word format on individual PCs. There is no centralised workflow system.
- NADRA VeriSys is used for CNIC verification, but only for shortlisted candidates and without system-level integration. HEC degree verification and domicile verification are fully manual.

The production environment is hosted on NTC National Data Center (Islamabad) as Virtual Dedicated Servers. A VPN connection between FPSC HQ and NTC is in place. The new system will be deployed on infrastructure hosted at NTC under a separate procurement.

2.3 Project Objectives

The primary objectives of project are to fully automate, implement and deploy:

- The **Enterprise Resource Planning (ERP) system for**
 - Digitalization of In-House Processes
 - Automation of Examination System
 - Development of **Question Databank (QDB)**
 - Implementation of **Computer-Based Testing (CBT)**

2.4 Stakeholder Roles

FPSC is the project owner. The Project Steering Committee (PSC), chaired by the Secretary/PAO, takes strategic decisions. The Project Director (PD) heads execution. A Project Management Cell (PMC) within FPSC IT Wing provides day-to-day coordination. FPSC provides access to information, stakeholders, premises, and timely decisions.

PISC (NUST) is the supervisory and implementation technical consultant. PISC reviews the Vendor's deliverables, validates milestone achievements, conducts quality assurance, and recommends acceptance to FPSC. Monitoring and Evaluation (MNE) of the project deliverables will be supervised by PISC being an independent consultant, and state owned organization engaged will report the implementation progress and deliverables to PISC. PISC does not have contractual authority over the Vendor. All approvals and payments flow from FPSC.

Software Vendor (selected through this RFP) designs, develops, tests, deploys, trains, and maintains the integrated software platform. The Vendor reports to PISC and {ISC coordinates with the PMC. The Vendor shall cooperate with PISC during technical reviews and quality assessments.

Hardware/Infrastructure Vendors and NTC are engaged under separate contracts. The Software Vendor shall coordinate with them during deployment but is not responsible for hardware procurement, network infrastructure, or data center services.

2.5 FPSC Organizational Structure

The Vendor shall engage with the following wings/directorates during BPR and requirements gathering. Each has distinct workflows that the system must automate:

Wing / Directorate	Relevance to System
Competitive Examination (CE) Wing	CSS lifecycle: advertisements, scheduling, hall management, attendance monitoring, result processing, personal hearings, viva voce, group/service allocation.
Recruitment Wing (T and S and FS Directorates)	GR lifecycle: requisitions, advertisements, test conduct, document scrutiny, shortlisting, interviews, nominations, alternate nominations.
Secrecy Wing	Question paper lifecycle: examiner management, paper preparation, printing, packing, answer book handling, fictitious numbering, OMR scanning, marks entry, result compilation.
Curriculum, Research, and Coordination Wing	Syllabus preparation for all exam types, difficulty level definition, subject expert coordination.
IT Wing	Online Recruitment System, database, website, admission

Wing / Directorate	Relevance to System
	certificates, roll numbers, data operations, system configuration.
HR Directorate	Duty rosters and deployment of invigilators and supervisory staff for examinations. Fee collection, examiner payments, budget tracking, financial reconciliation, TA/DA claims of examination staff.
Logistics Directorate	Procurement and distribution of examination material, transport arrangements for exam operations.
Psychology Directorate	Psychological assessments for CSS candidates, scheduling and reporting.
Program Management Section	Interview scheduling, viva voce logistics, Commission member coordination.

3. Scope of Work

3.1 Overview

The vendor shall design, develop, test, deploy, and maintain an ERP solution for digitalization of in-house processes, and automation of examination processes. The platform shall automate FPSC's recruitment and examination operations and establish a Computer-Based Testing capability backed by a secure Question Data Bank.

The platform manages the entire lifecycle of recruitment at FPSC from the moment a requisition arrives to the day a candidate receives a nomination letter. Every step in between, whether it is advertising a post, receiving applications, conducting an examination, compiling results, scrutinizing documents, scheduling interviews, or issuing correspondence, happens within the same system. All wings and directorates work on the same data in real time. The modules described in Sections 3.2 through 3.7 organize these capabilities by function.

The system shall also be extendable to future requirements including Computer Adaptive Testing, internet-based proctored testing, and AI-augmented evaluation. These are not in scope for this RFP but the architecture must not preclude them.

Business Process Re-engineering (BPR): The functionalities listed in Sections 3.2 through 3.7 represent the preliminary scope as defined in the approved project documentation. The Vendor shall conduct a comprehensive BPR exercise as the first phase of the project, engaging with all FPSC wings and directorates to map existing workflows, identify redundancies, and produce a detailed Software Requirements Specification (SRS). The SRS, once approved by FPSC through PISC review, shall become the binding requirements baseline for development.

Traceability Requirement: The SRS produced by the Vendor shall include a traceability matrix mapping all workflows across all FPSC wings and directorates identified in Section 2.5 to the corresponding system functionality, responsible role/wing, and workflow state. Any existing activity proposed for elimination or re-engineering during BPR shall be explicitly justified and approved by FPSC.

Compliance Matrix: Bidders shall respond to each functionality listed below in the Compliance Matrix (Annexure A) with one of: Fully Compliant / Partially Compliant (with explanation) / Non-Compliant. This response demonstrates the Vendor's understanding of scope and is evaluated under Section 9 (Technical Evaluation).

The six modules constitute the contractual scope however all functional requirements listed in section 3 and Annex-A may be considered minimum scope which can be revised during BPR and SRS finalization stage without any additional cost. The table below maps the modules to the enterprise capability workstreams they collectively deliver. The modules and the delivery phases in Section 7 remain the basis for contracting, milestones, and acceptance.

Workstream	Executive Purpose	Existing RFP Coverage
ERP Core Platform	Establish centralized enterprise-wide examination and recruitment platform	Sections 3 and 4
General Recruitment System	Automate recruitment lifecycle and candidate processing	Section 3.2
Competitive Examination System	Digitize competitive examination operations and workflows	Section 3.3

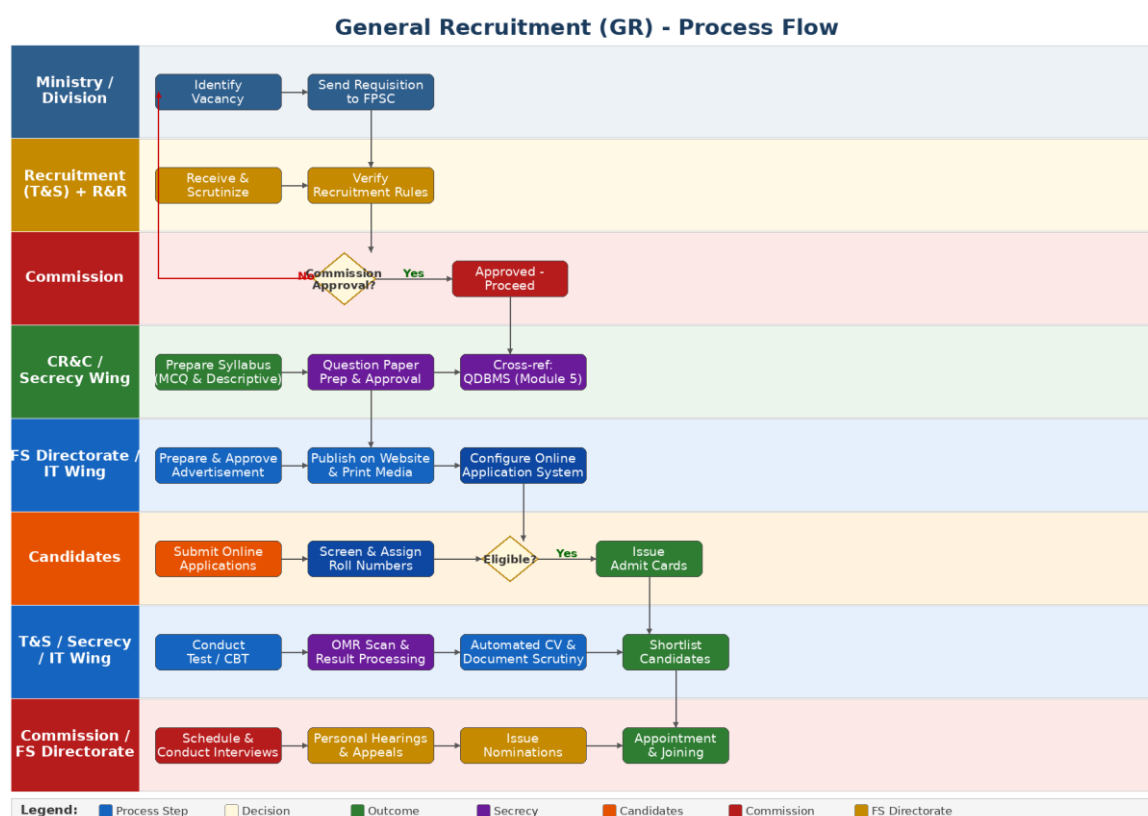
QDBMS & CBT Platform	Enable secure digital question management and CBT operations	Section 3.6
Workflow & eCase Management	Reduce inter-departmental delays through digital approvals and case workflows	GR-1.12, CE-2.10, Cross-Cutting Services
Identity & Access Management	Ensure secure user authentication and role-based access control	Sections 3.8 and 4.9
Integration & API Gateway	Enable interoperability with external government and enterprise systems	Section 3.9
Analytics & Decision Support	Provide executive dashboards, reporting, and operational analytics	SUP-6.6, TAB Modules
Notification & Communication Services	Centralize candidate and operational communication services	CC-02, MOB-04
Mobile & Tablet Ecosystem	Extend services to mobile and tablet-based operations	Section 3.10
Security & Compliance Services	Ensure auditability, governance, monitoring, and examination security	Sections 3.6, 3.8, 4.7, 4.8, 4.9
DR & Infrastructure Automation	Ensure operational continuity and disaster resilience	Section 4.7
DevSecOps & CI/CD	Support controlled deployment, release, and operational automation	Section 5 – Non-Functional Requirements
Monitoring & SIEM	Enable centralized monitoring and operational visibility	SUP Modules and Monitoring References
Data Migration & Legacy Modernization	Support migration from legacy FPSC systems and data consolidation	Sections 3.9 and 4.8

Table: Module-to-Enterprise-Workstream Coverage Map

3.2 Module 1: General Recruitment (GR)

This module automates the recruitment lifecycle for ex-cadre posts (BPS-16 and above) advertised through FPSC's consolidated advertisements. The workflow spans from requisition receipt through final nomination, involving 10 wings/directorates and the Commission across 9 sequential phases and approximately 80 distinct activities.

The process flow below illustrates the end-to-end GR lifecycle. Color coding indicates the responsible FPSC wing/directorate at each step.



The table below specifies the system capabilities required for each phase in sequential order. GR-1.1 is the starting point of the recruitment cycle (requisition receipt) and GR-1.9 is the endpoint (final nomination and follow-up). Each row represents a phase that follows the preceding one in the standard workflow. GR-1.10 through GR-1.13 provide supporting functions that operate across all phases. During BPR, the Vendor shall map these phases to the detailed activity-level workflows of each wing/directorate.

Ref	Functionality	Responsible Wing
GR-1.1	Requisition receipt, registration, and tracking. Scrutiny against roster, domicile, and recruitment rules (R and R verification). Auto-flagging of mismatches. Submission to Commission for approval. On approval, auto-trigger of syllabus preparation workflow to CR and C Wing.	T and S, R and R, Commission, CR and C
GR-1.2	Syllabus preparation (MCQ and descriptive) by CR and C Wing with Commission approval. Examiner selection, question paper preparation, scrutiny, and approval by Secrecy Wing. Cross-reference: Module 5 (QDBMS).	CR and C, Secrecy, Commission
GR-1.3	Consolidated advertisement preparation from approved requisitions (including re-advertisements). FS Directorate approval. Publication on website and print media. Configuration of online application system. Uploading of syllabi.	FS, IT
GR-1.4	Registration of applicant online profile (personal data, education, experience, fee). Online application submission.	IT, T and S, Commission

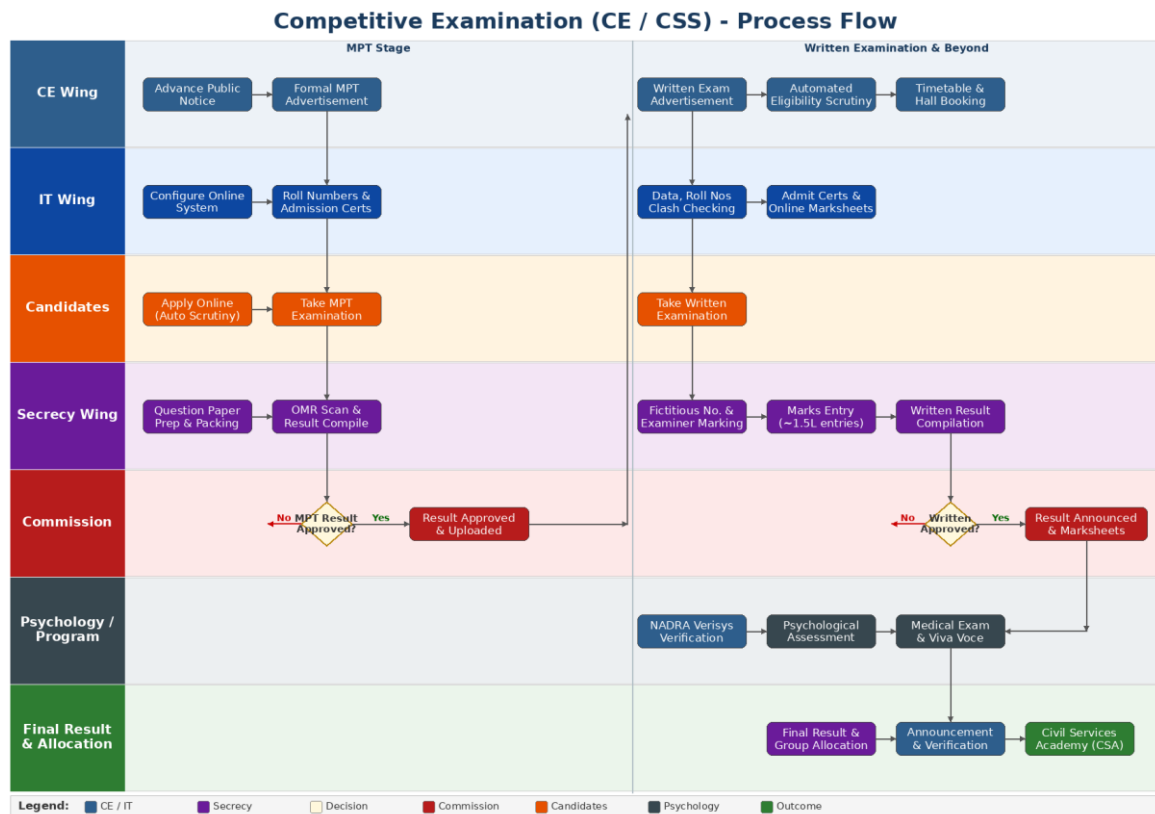
Ref	Functionality	Responsible Wing
	Data cleaning, migration to local server, provision to T and S and Facilitation Centre. Commission approval for manual applications. Roll number assignment. Centre/case-wise figures. Identification of non-test and fast-track cases.	
GR-1.5	Timetable preparation and approval. Hall/sub-centre booking. Case allocation among halls. Circulation of breakup to IT, Secrecy, Logistics, HR. Question paper printing, counting, bundling, packing. Sub-centre candidate allocation. Admission certificates online. Attendance sheets. SMS/email intimations. Duty lists (cross-ref: Module 6). Material distribution (cross-ref: Module 6). Test conduct and attendance recording.	T and S, Secrecy, HR, Logistics, IT
GR-1.6	Attendance marked in database. Receipt, opening, counting of answer sheet parcels. OMR scanning. Manual re-verification of 5% OMR sheets. Result compilation. Commission approval. Result upload, merit list, marksheet generation.	Secrecy, Commission, IT
GR-1.7	Public notice for interview. Scheduling of candidates for document scrutiny. Automated scrutiny of candidate documents and CVs against advertised conditions (education, experience, age, domicile, quota, CNIC) using configurable rule engine with CV parsing, weighted scoring, ranked shortlist generation. Officer review, override, audit trail. Shortlisting/rejection with Commission approval.	T and S, Program, Commission, IT
GR-1.8	Registration of subject specialists and interview panel members with automated CV scrutiny against panel eligibility criteria. Draft and fair summary bio-data generation. Interview call letters. Placement of interview programs on website. Conduct of interviews and personal hearings of rejected candidates. Appeals processing by FS Directorate.	FS, IT, Program, Commission
GR-1.9	Commission approval for nominations. Nomination letters to selected candidates and intimation to requisitioning department. Website upload and database recording. Tracking of joiners/non-joiners. Alternate nominations from reserved merit list (within one year) with Commission approval. Follow-up and monitoring.	FS, Commission, IT
GR-1.10	Recruitment rules registration, search, and comparative statement sheet generation.	R and R
GR-1.11	Quota roster registration, search, and vacant post list for circulation (jobs planner).	T and S
GR-1.12	Interactive eCase flow management across all phases: eCase files, workflows, attachments, noting, actions, search/trail, and statistics.	All Wings
GR-1.13	Intimations to candidates (SMS, email, push notifications) on major milestones. Generation of all correspondence using configurable templates. Ministry/Division/candidate case tracking. Biometric capturing and online fee payment. Historic data migration. FPSC Annual Report integration.	IT, All Wings

***Note: Mobile and tablet application requirements are specified separately in Section 3.10.**

3.3 Module 2: Competitive Examination (CE)

This module automates the CSS Competitive Examination lifecycle, including the Mandatory Preliminary Test (MPT), written descriptive examination, psychological assessment, medical examination, viva voce, and group/service allocation. The workflow spans 9 sequential phases and approximately 149 distinct activities across CE Wing, Secrecy Wing, IT Wing, Psychology Directorate, Program Management, HR, Admin/Logistics, Finance, and the Commission.

The process flow below illustrates the end-to-end CSS examination lifecycle.



The table below specifies the system capabilities required for each phase in sequential order. CE-2.1 is the starting point (examiner panel and question paper preparation) and CE-2.9 is the endpoint (final result, group/service allocation, and forwarding to Establishment Division). Each row represents a phase that follows the preceding one in the standard CSS examination workflow. CE-2.10 provides supporting functions that operate across all phases

Ref	Functionality	Responsible Wing
CE-2.1	Examiner panel preparation, Commission approval, confirmation for question paper setting. Preparation, dispatch, receipt, typing, comparing, and proof reading of question papers. Cross-reference: Module 5 (QDBMS).	Secrecy, Commission
CE-2.2	Advance public notice and formal advertisement for MPT. Configuration and testing of online system. Receipt and monitoring of online applications. Registration of applicant profile (personal data, education, experience, subject/group selection, fee,	CE, IT, Candidate

Ref	Functionality	Responsible Wing
	documents) with automated scrutiny module for eligibility verification.	
CE-2.3	Data cleaning, roll number assignment, centre-wise figures. Timetable, hall booking, duty lists. Admission certificates, attendance sheets online. Question paper printing, packing, distribution. Exam folder preparation, material distribution, staff training. Conduct of MPT, attendance monitoring, parcel dispatch.	IT, CE, Secrecy, HR, Admin/Logistics
CE-2.4	Separation of absent candidates. OMR scanning, 5% manual verification. MPT result preparation. Commission approval. Result upload on website.	Secrecy, CE, Commission, IT
CE-2.5	Written exam advertisement (MPT-qualified only). Hardcopy receipt, TR processing, data feeding and verification, fresh roll numbers. Eligibility reports. Automated eligibility scrutiny using configurable rule engine. Rejection preparation, Commission approval, personal hearings, rejection letters, website upload of rejected list.	CE, Admin, Finance, IT, Commission
CE-2.6	Written exam timetable, clash checking, hall booking. Centre-wise/subject-wise breakup and circulation. Transport and security arrangements. Admission certificates, attendance sheets, exam material provision, duty lists. Question paper preparation and distribution. Exam folder preparation, staff training. Conduct of written examination. Attendance monitoring and database marking.	CE, IT, Secrecy, Admin, HR
CE-2.7	Receipt of parcels. Fictitious roll numbering process (register, affixing, verification, cutting originals). Answer book sorting and packing. Delivery to examiners. OMR scanning. Detailed scrutiny of appeared candidates (Commission approval). 5% OMR verification. Answer book scrutiny, discrepancy resolution. Marks entry (approximately 1.5 lakh entries). Result compilation, committee checking, Commission approval. Result announcement, online marksheets.	Secrecy, CE, Commission, IT
CE-2.8	NADRA Verisys verification (CNIC, domicile, religious status). Scheduling for psychological assessment and viva voce. Conduct of psychological assessment with document verification. Medical examination arrangements. Summary bio-data preparation and authentication. Conduct of viva voce.	CE, Psychology, IT, Program
CE-2.9	Re-counting requests processing. Medical suitability scrutiny for group allocation. Vacancy calculation and distribution (merit, provincial, women, minorities quotas). Commission and Establishment Division approval. Public notice of vacancies. Final result preparation after viva voce. Group/Service preference compilation and allocation (merit + domicile + preference + suitability). Result announcement. Forwarding to Establishment Division. Re-allocation for non-joiners. Database incorporation.	Secrecy, CE, Commission, IT
CE-2.10	Automated data migration between production and local system. Interactive eCase flow management. Vacant seat management and quota circulations. Intimations to candidates (SMS, email, push). Correspondence generation using configurable templates. Candidate case tracking. Historic data migration. FPSC Annual Report integration. Biometric capturing and online fee payment.	IT, CE, All Wings

***Note: Mobile and tablet application requirements are specified separately in Section 3.10.**

3.4 Module 3: Unified Examination Module (UEM)

This module provides a configurable examination engine for all other examination types conducted by FPSC: FPOE, SOPE, Civil Judges, AD Survey of Pakistan, Gilgit-Baltistan Competitive Examinations, internal recruitments, psychological assessments, situational tests, and any additional types defined in the future. The module must be configurable to accommodate new exam types without code changes.

Because UEM is a generic, configurable engine, it follows the same phased workflow pattern as GR (Module 1) and CE (Module 2) but with administrator-definable stages. Where an exam type follows the GR or CE pattern, UEM shall re-use the same workflow engine with exam-type-specific configuration rather than separate code.

Ref	Functionality	Responsible Wing
UEM-3.1	Exam type generator: FPOE, SOPE, Civil Judges, AD Survey, GB Competitive Examinations, internal recruitments, psychological assessments, situational tests, and additional types definable by administrator without code changes.	IT/Admin
UEM-3.2	Requisitions registration, search, statistics, and current status tracking.	Recruitment
UEM-3.3	Candidate profile management (profile-based and non-profile applications). Online application submission. Candidate dashboard with progress tracking.	IT / Candidate
UEM-3.4	Advertisement consolidation. Recruitment rules and quota roster management. Migration to production server.	FS / Relevant Wing, R and R
UEM-3.5	Pre-exam reports: centre/batch-wise statements, timetable, hall allocation, admission certificates, attendance sheets. Automated data migration between production and local system.	IT, Relevant Wing
UEM-3.6	Online document submission for shortlisting with automated scrutiny module. Automated CV/resume evaluation with configurable rules: document parsing, field extraction, weighted scoring, ranked shortlist generation.	Relevant Wing
UEM-3.7	Marksheet generation. Recounting requests. Scheduling of psychometric, medical, viva voce examinations. Personal hearings. Final merit list and re-allocation on unfilled seats.	IT, Secrecy, Program
UEM-3.8	Intimations (SMS, email, push). Correspondence using templates. Case tracking. Historic data migration. FPSC Annual Report integration. Training walkthrough. Biometric capturing and online fee payment.	IT, All Wings

3.5 Module 4: Official Website and Candidate Portal

The Vendor shall develop an official website and integrated candidate portal replacing the existing FPSC website. The portal shall serve as the single point of interaction for candidates and the public.

Ref	Functionality
WEB-4.1	Industry-standard CMS enabling FPSC staff to manage website content without developer intervention.

Ref	Functionality
WEB-4.2	Mobile-responsive design compatible with all major mobile browsers.
WEB-4.3	Content accessibility including provisions for persons with special needs.
WEB-4.4	Search engine optimization (SEO).
WEB-4.5	Well-planned information architecture for intuitive navigation.
WEB-4.6	Cross-browser compatibility and consistency.
WEB-4.7	Effective navigation: search, multiple exploration paths, custom 404, internal linking, informative header/footer.
WEB-4.8	Valid markup, clean code, and proper error handling.
WEB-4.9	Contrasting color scheme meeting accessibility guidelines.
WEB-4.10	Google Analytics and Search Console integration.
WEB-4.11	Usable forms with validation and user-friendly error messages.
WEB-4.12	Social media links integration.
WEB-4.13	Staff login with role-based access to backend content management.

3.6 Module 5: CBT and Question Data Bank

This module comprises of two tightly integrated components: the computer-based testing (CBT) system for secure exam delivery, and the Question Data Bank Management System (QDBMS) for question lifecycle management. Together, they replace the current paper-based examination process.

5A. Question Data Bank Management System (QDBMS)

Ref	Functionality
QDB-5.1	Comprehensive question data bank with capacity for one million items, tagged with subject, topic, difficulty level, usage count, question age, and configurable metadata.
QDB-5.2	Question authoring: MCQ (single/multi-answer), true/false, matching, fill-in-the-blank, with future extensibility to essay/subjective types.
QDB-5.3	Dynamic, hierarchical classification taxonomy configurable by administrator without code changes.
QDB-5.4	Question versioning with full change history and rollback capability.
QDB-5.5	Status workflow: Draft, Under Review, Approved, Active, Retired. Only Approved/Active questions available for paper generation.

Ref	Functionality
QDB-5.6	Role-based access: authors, reviewers, approvers. No single role can view a complete exam paper.
QDB-5.7	Dual-authorization required for question export or paper generation.
QDB-5.8	Automated paper generation: randomized assembly with configurable constraints (topic coverage, difficulty distribution, reuse limits).
QDB-5.9	Psychometric analytics after each exam: facility index, discrimination index, discrimination efficiency, reliability coefficient.
QDB-5.10	Automatic flagging of questions requiring revision based on psychometric thresholds.
QDB-5.11	Bulk import from standard formats (Excel, CSV).
QDB-5.12	Per-question audit trail: created by, reviewed by, approved by, used in which exam, when, and all modifications.
QDB-5.13	Database encryption at rest (AES-256) and in transit (TLS 1.3).

Note: The Vendor provides the question bank platform and authoring tools only. The one million question items are developed by subject matter experts engaged separately by FPSC and are not part of this RFP.

5B. Computer-Based Testing (CBT) System

Ref	Functionality
CBT-5.14	CBT software with user management (authentication, roles) and session management (start, pause, resume, auto-submit).
CBT-5.15	Safe Exam Browser (SEB) integration or equivalent lockdown browser.
CBT-5.16	Question randomization: question order, option order, pool-based selection ensuring no two adjacent candidates see the same sequence.
CBT-5.17	Configurable timer with warnings and auto-submission on expiry.
CBT-5.18	Biometric candidate verification integrated with NADRA at the exam site.
CBT-5.19	Live invigilator dashboard: real-time status, progress, time remaining, anomaly flags.
CBT-5.20	Offline-capable exam delivery on local edge server. Encrypted exam packages distributed before exam day, decryption keys released at exam time.
CBT-5.21	Multi-shift scheduling with secure data wipe between shifts.
CBT-5.22	Device swap: candidate switches terminal mid-exam without data loss.
CBT-5.23	Post-exam encrypted result sync with integrity verification.
CBT-5.24	Candidate interface: intuitive, accessible, question flagging, navigation panel, review screen.
CBT-5.25	Exam package encryption (AES-256); decryption key via separate secure channel.
CBT-5.26	Extensibility to subjective/essay-type questions without architectural changes.

3.7 Module 6: Supporting Module

This module provides operational support functions integrated with the examination workflow for resource planning, logistics, and management oversight.

Ref	Functionality
SUP-6.1	HR and duty deployment: invigilator/supervisor/staff assignment integrated with leave management.
SUP-6.2	Examination inventory management: track materials issued to and received from centers.
SUP-6.3	Exam transport management: material dispatch and staff deployment logistics.
SUP-6.4	Attendance and leave management linked to exam resource planning.
SUP-6.5	Library management module.
SUP-6.6	Decision support system: dashboards, analytics, configurable KPIs, real-time status, trend analysis.
SUP-6.7	Psychometric testing support: scheduling, administration, and reporting for CSS psychological assessments.

3.8 Cross-Cutting Requirements

The following requirements apply across all modules. Where an individual module table refers to one of these functions, this section is the authoritative specification for it.

Ref	Functionality
CC-01	Comprehensive audit trail across all modules: every create, read, update, and delete action logged with user identity, timestamp, IP address, and before/after values. Audit logs must be tamper-evident.
CC-02	Notification engine: SMS, email, in-app, and push notifications triggered by configurable workflow events. Delivery confirmation and retry logic required.
CC-03	Reporting engine: standard reports (exam-wise, centre-wise, subject-wise, quota-wise) plus an ad-hoc report builder enabling FPSC staff to create custom reports without developer intervention. All reports are exportable to PDF, Excel, and CSV.
CC-04	Template engine: all correspondence, letters, certificates, and reports can be generated from configurable templates that FPSC staff can modify.
CC-05	Full-text search across candidates, requisitions, cases, questions, and all system entities.
CC-06	Configurable CV/resume scrutiny service: a shared, reusable system service invocable by any module (GR, CE, UEM, panel/specialist registration). Administrators shall define custom scrutiny criteria per post/exam (rule sets, weights, knockout conditions, preferred keywords, minimum thresholds) without code changes. The service shall parse uploaded CVs and testimonials (PDF, DOCX, scanned images), extract structured fields (qualifications, institutions, experience, certifications), score candidates against the defined criteria, and produce ranked shortlists with transparent scoring rationale. Officer override and audit logging are mandatory.
CC-07	Grievance and complaint management: candidates shall be able to submit grievances, representations, and review petitions through the candidate portal. The system shall assign each submission a unique tracking number, route it to the responsible wing/directorate based on configurable rules, track status through receipt, examination, personal hearing (if applicable), and final decision. Decision recording shall capture the outcome, deciding authority, date, and rationale. Candidates shall receive status updates

Ref	Functionality
	at each stage via SMS, email, and portal notification. The system shall support scheduling of personal hearings for rejected candidates with calendar integration to panel/Commission member availability. All grievance actions shall be logged in the audit trail. Reporting shall include: grievances received, pending, resolved, average resolution time, and escalation status.

3.9 External System Integration

The platform shall integrate with the following external systems and services. The Vendor is responsible for the software integration (API development, data mapping, error handling, and testing). Network connectivity, VPN tunnels, and third-party service agreements are provided by FPSC or the respective agency under separate arrangements.

Ref	External System	Integration Scope
INT-01	FPSC Online Recruitment System (Oracle 11g / MySQL)	Full data migration of 17 years of historical data (approximately 100 GB) into the new platform. The existing system will be decommissioned after successful migration and reconciliation sign-off. Migration requirements are detailed in Section 5.6.
INT-02	NADRA VeriSys	Real-time CNIC verification for candidate identity at registration and biometric verification at CBT sites. Integration via NADRA's secure API over a dedicated VPN tunnel.
INT-03	HEC (Higher Education Commission)	Degree and qualification verification. Integration method (API, batch file, or manual portal) to be determined during BPR based on HEC's available interfaces. If no automated interface is available from HEC, the system shall support manual verification entry with audit trail.
INT-04	Payment Gateway	Online fee collection for examination applications. Integration with a State Bank of Pakistan approved payment gateway supporting credit/debit cards, internet banking, and mobile wallets.
INT-05	SMS Gateway	Bulk SMS delivery for candidate notifications, OTP authentication, and exam-day alerts. Integration via standard SMS API with delivery status tracking.
INT-06	Email Service	Transactional and bulk email for candidate correspondence, admission certificates, result notifications, and system alerts.

During BPR, the Vendor shall identify any additional integration points required by FPSC's operational workflows and propose them in the SRS for FPSC approval.

The table below sets out, for each interoperability area, the business objective and the corresponding requirement reference.

Integration Area	Business Objective	How Current RFP Addresses It
Legacy System Integration	Ensure continuity and migration from existing FPSC systems	Migration and integration requirements included in Sections 3.9 and 4.8
NADRA Integration	Enable automated candidate identity verification	External integration framework defined in Section 3.9

HEC Verification	Automate degree and credential validation	Verification services referenced in Section 3.9
Payment Gateway Integration	Enable digital fee collection and reconciliation	Payment and transaction workflows included in Sections 3.2 and 3.9
Communication Services	Improve candidate communication and operational responsiveness	Notification engine and mobile communication services in CC-02 and MOB-04
QDBMS & CBT Integration	Enable secure end-to-end digital examination operations	Integrated QDBMS and CBT architecture defined in Section 3.6
Edge Examination Synchronization	Support nationwide secure CBT operations	Synchronization architecture and edge-node operations in Sections 4.6 and 4.7
Mobile & Tablet Ecosystem	Extend operational services to mobile and tablet platforms	Mobile and tablet integration framework defined in Section 3.10
Analytics & Executive Reporting	Enable centralized monitoring and decision support	DSS and executive dashboard services included in SUP-6.6 and TAB modules
Unified Enterprise Platform	Establish centralized enterprise-wide examination management	Centralized EMS and architecture framework defined in Sections 3 and 4

Table: Interoperability Matrix

3.10 Mobile and Tablet Applications

In addition to the mobile-responsive web portal, the Vendor shall develop native or cross-platform mobile applications for the following user groups. The mobile applications shall share the same backend, the same database, and the same authentication system as the web platform. They are extensions of the integrated system, not standalone applications.

3.10.1 Candidate Mobile Application (Android and iOS)

A mobile application for candidates providing the following functionality:

Ref	Functionality
MOB-01	Candidate registration, profile creation, and document upload (photographs, certificates, domicile, CNIC) directly from the phone camera.
MOB-02	Online application submission for advertised posts/examinations with fee payment through mobile banking, JazzCash, EasyPaisa, or other supported payment methods.
MOB-03	Admission certificate viewing and download.
MOB-04	Real-time push notifications for application status updates, admit card availability, exam schedule, result announcements, interview calls, and nomination status.
MOB-05	Result and marksheet viewing.
MOB-06	Grievance and representation submission with tracking (as specified in CC-07).
MOB-07	Centre and seat information with map integration for exam venue location.
MOB-08	Candidate profile management: update personal details, change password, and manage notification preferences.

Note: The candidate mobile application is for registration, tracking, and information only. CBT examinations shall not be conducted on mobile devices; they require SEB-locked desktops at designated CBT sites.

3.10.2 Executive Dashboard Application (Tablet, Android and iOS)

A tablet-optimized application for FPSC senior officers, Commission members, and the Project Director, providing decision-support information at a glance:

Ref	Functionality
TAB-01	Real-time recruitment cycle dashboard: cases in progress by wing, pending approvals, bottlenecks, and SLA compliance indicators.
TAB-02	Examination dashboard: upcoming exams, site readiness status, live exam monitoring (candidate check-in progress, exam completion rates), and post-exam result summary.
TAB-03	Key performance indicators (KPIs) aligned with the project objectives in Section 2.3: recruitment cycle duration, question paper generation time, result announcement time, and process automation percentage.
TAB-04	Approval workflow: pending items requiring Commission or senior officer approval, with the ability to review supporting documents and record approval or rejection decisions directly from the tablet.
TAB-05	Notification centre: consolidated view of system alerts, milestone updates, and escalations.
TAB-06	Secure access with multi-factor authentication. Offline viewing of the most recently synced dashboard data when network is unavailable.

3.10.3 Invigilator Application (Tablet, Android and iOS)

A tablet application for invigilators and supervisory staff at CBT sites, replacing the need for a dedicated desktop for monitoring duties:

Ref	Functionality
INV-01	Candidate check-in: scan candidate CNIC or admission certificate barcode/QR code, trigger biometric verification, and record attendance.
INV-02	Live exam monitoring dashboard: candidate progress (current question, time remaining, completion status), anomaly flags, and seat map view.
INV-03	Incident logging: record terminal failures, candidate complaints, identity verification issues, and time extension approvals with timestamp and invigilator ID.
INV-04	Attendance report: real-time count of present, absent, and late candidates.
INV-05	Post-exam checklist: confirm candidate departure, verify terminal lockdown, confirm local cache wipe status (for edge sites), and submit site closure report.
INV-06	Secure access with role-based authentication. The application shall only function within the CBT site network (geofenced or network-restricted).

Exclusions: The following functions shall remain desktop/web-only and are not required on mobile or tablet: QDBMS (question authoring, review, approval, paper generation), full EMS administrative operations (requisition processing, advertisement preparation, result

compilation), CBT exam delivery (requires SEB-locked desktops), and system administration.

4. Deployment Architecture

4.1 Architecture Overview

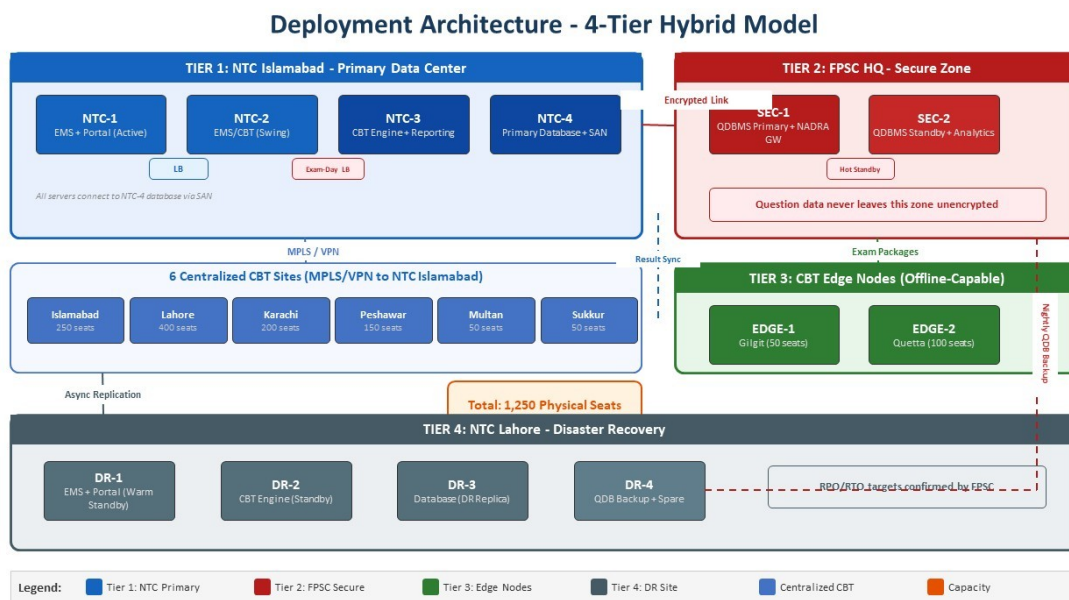
The system shall be deployed on a 4-tier hybrid architecture combining centralized cloud hosting at NTC with on-premise secure processing at FPSC HQ and local edge servers at connectivity-challenged CBT sites. This architecture balances operational efficiency (centralized management), data confidentiality (question data isolated on-premise), and exam resilience (offline-capable edge nodes).

The four tiers are:

- Tier 1, NTC Islamabad (Primary Data Center): hosts EMS, Portal, CBT central engine, and primary database.
- Tier 2, FPSC HQ Secure Zone: hosts QDBMS, paper generation engine, and NADRA integration gateway.
- Tier 3, CBT Edge Nodes: dedicated local exam servers at Gilgit and Quetta for offline-capable exam delivery.
- Tier 4, NTC Lahore (Disaster Recovery): geo-redundant replica of Tier 1 and Tier 2 data.

The remaining six CBT sites (Islamabad, Lahore, Karachi, Peshawar, Multan, Sukkur) operate in centralized mode, where candidate desktops connect to the CBT engine at NTC via MPLS/VPN. The software must be designed to support deploying additional edge servers at any site in the future without code changes.

The Vendor's proposed software architecture must operate within this deployment model.



4.2 Tier 1: NTC Islamabad (Primary Data Centre)

Four servers co-located at NTC Islamabad, running on a virtualized HCI stack. The Vendor deploys application software on virtual machines provisioned on this infrastructure.

Server	Role	Description
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Server	Role	Description
NTC-1	EMS + Portal (Active)	Hosts EMS modules (GR, CE, UEM, Supporting), candidate portal, website, and notification services. Load-balanced with NTC-2 during normal operations. Exam-day mode: runs EMS/Portal alone while NTC-2 switches to CBT duty.
NTC-2	EMS + Portal / CBT Swing	Load-balanced pair with NTC-1 for EMS/Portal during normal operations. Exam-day mode: switches to CBT Engine duty, forming a load-balanced CBT pair with NTC-3. EMS/Portal runs on NTC-1 alone (sufficient for low exam-day portal traffic). This dual-role design provides CBT redundancy without additional servers.
NTC-3	CBT Central Engine + Reporting	Manages exam orchestration for the 6 centralized CBT sites: exam session management, question delivery, answer collection, invigilator dashboards, real-time monitoring. Post-exam: result compilation, reporting, and analytics. Exam-day mode: load-balanced with NTC-2 for CBT delivery. If either server fails, the surviving server handles the full CBT load.
NTC-4	Primary Database	Hosts EMS transactional database, reporting database, and audit logs. Connected to SAN storage for high-performance I/O. Replicates asynchronously to DR-3 at NTC Lahore.

Vendor software requirement: The CBT application must be designed as a stateless, horizontally scalable service that can run on multiple nodes simultaneously. Session state must be managed externally (database or distributed cache) so that any node can serve any candidate request. This enables the exam-day workload switchover between NTC-2 and NTC-3 without session loss.

4.3 Tier 2: FPSC HQ Secure Zone

Two servers located at FPSC Headquarters in a physically secured environment. This tier handles all question-related operations and NADRA integration. Question data never leaves this zone in unencrypted form.

Server	Role	Description
SEC-1	QDBMS Primary	Runs QDBMS application and database on the same server. Also runs: exam package builder (generates encrypted CBT packages from QDB content), NADRA integration gateway (secure VPN endpoint for identity verification). Accessible only from FPSC internal network and via dedicated encrypted link to Tier 1. No public internet exposure.
SEC-2	QDBMS Standby	Hot standby mirror of SEC-1. Takes over if SEC-1 fails. During normal operations, serves as the read-only analytics node for psychometric reports and question bank statistics, offloading SEC-1. Sends nightly encrypted QDB backup snapshots to DR-4 at NTC Lahore.

Vendor software requirement: The QDBMS must enforce split-knowledge access control; no single user role can view or assemble a complete exam paper. Paper generation requires dual authorization from two separate role holders. The exam package builder must operate

in a network-isolated mode during paper generation (no outbound connections). All QDB database content must be encrypted at rest using AES-256.

4.4 Tier 3: CBT Edge Nodes

Two dedicated edge servers deployed at Gilgit and Quetta, where network connectivity to NTC Islamabad is less reliable. These servers provide offline-capable exam delivery, ensuring exams continue uninterrupted even if the WAN link drops during the exam window.

Server	Location	Seats	Description
EDGE-1	Gilgit	50	Receives encrypted exam package from Tier 2 before exam day. Runs CBT locally during exam window; SEB terminals connect to EDGE-1 only. If WAN link is available, streams monitoring data to NTC-3 in real time. If WAN link drops, exam continues uninterrupted. Results sync post-exam.
EDGE-2	Quetta	100	Same architecture as EDGE-1. Handles 100 concurrent candidates locally. Post-exam: encrypted results synced to NTC-3 with integrity verification (checksum).

The remaining six CBT sites operate in centralized mode:

Site	Seats	Connectivity
Islamabad (FPSC HQ)	250	MPLS to NTC Islamabad. Same city; lowest latency, highest reliability.
Lahore	400	MPLS to NTC Islamabad. NTC DR site also in Lahore; automatic failover path available.
Karachi	200	MPLS to NTC Islamabad. Major NTC backbone node; reliable connectivity.
Peshawar	150	MPLS to NTC Islamabad. Provincial capital with established NTC infrastructure.
Multan	50	MPLS to NTC Islamabad. Regional office with standard connectivity.
Sukkur	50	MPLS to NTC Islamabad. Regional office with standard connectivity.

Total physical seats across all 8 sites: 1,250. The system shall be designed and load-tested for 3,000 concurrent CBT users to support multi-shift operations and future expansion.

Vendor software requirement: The CBT software must support both centralized mode (terminals connect to NTC) and edge mode (terminals connect to local server) using the same codebase. Switching a site from centralized to edge mode shall require only configuration changes and deployment of the edge server package, with no code modifications. This ensures FPSC can add edge servers at additional sites in the future based on operational experience.

4.5 Tier 4: NTC Lahore (Disaster Recovery)

Four servers co-located at NTC Lahore, providing geo-redundant disaster recovery for Tier 1 and backup storage for Tier 2.

Server	Role	Description
DR-1	EMS + Portal (Warm Standby)	Replica of NTC-1/NTC-2. Receives asynchronous state replication. Activatable within RTO.
DR-2	CBT Engine + Reporting (Standby)	Replica of NTC-3. Can take over exam orchestration if NTC Islamabad is completely unavailable.
DR-3	Database (DR Replica)	Asynchronous replica of NTC-4. Connected to SAN storage unit at NTC Lahore.
DR-4	QDB Backup + Spare	Receives nightly encrypted QDB snapshots from SEC-2. Does NOT run an active QDBMS instance; question authoring only occurs in the Tier 2 secure zone. Also serves as a spare server reassignable if any DR node fails.

RPO and RTO targets for the production EMS database are specified in BR-06 (Section 5.5). The Vendor's software architecture must support asynchronous database replication to the DR site and must be capable of resuming operations from the DR site within the agreed RTO after a failover event.

4.6 Recruitment and Examination Operations Cycle

The recruitment and examination process operates as a continuous cycle. It begins with requisition processing and candidate registration, moves through examination preparation and conduct, and continues into result processing, document scrutiny, interviews, nominations, and post-cycle closure. The three phases below trace this cycle from preparation through execution to completion, covering both the examination operations and the administrative processing that follows.

4.6.1 Pre-Examination and Readiness Preparation

Pre-examination preparation begins well before the exam date and covers everything from question paper assembly to readiness confirmation at every CBT site. The objective of this phase is to ensure that all examination material is securely prepared, encrypted, distributed, and verified before a single candidate enters the exam hall.

Timing	Activity	System Actions and Responsible Parties
T minus 30 days	Syllabus and Scope Finalization	CR and C Wing finalizes the syllabus and question distribution requirements (number of questions, topic coverage, difficulty distribution) for the upcoming examination. This information is entered into the QDBMS as paper generation constraints.
T minus 21 days	Examiner Assignment and Question Readiness	Secrecy Wing verifies that sufficient Approved/Active questions exist in the QDB for the required subjects and difficulty levels. If gaps exist, question authoring and review workflows are triggered. The system shall flag any subject or topic where the available question pool is below the configured minimum threshold.

Timing	Activity	System Actions and Responsible Parties
T minus 7 days	Paper Generation and Package Assembly	Two authorized role holders (dual authorization) initiate paper generation on SEC-1. During assembly, SEC-1 enters network-isolated mode: all outbound connections are disabled. Questions are selected from the QDB per the configured constraints (topic coverage, difficulty distribution, reuse limits). The assembled paper is encrypted immediately using AES-256. The encrypted exam package is then distributed: for centralized sites (Islamabad, Lahore, Karachi, Peshawar, Multan, Sukkur), the package is uploaded to NTC-3. For edge sites (Gilgit, Quetta), the package is pushed to EDGE-1 and EDGE-2 over a secure channel. The decryption key is not transmitted with the package; it is held separately and released only at exam time through a different channel.
T minus 3 days	Site Readiness Verification	The system runs automated readiness checks at each CBT site. For centralized sites: MPLS connectivity to NTC is tested, bandwidth is measured, and latency is confirmed within acceptable thresholds. For edge sites: EDGE-1 and EDGE-2 confirm package receipt and checksum validation. All candidate desktops are verified: SEB is installed and responsive, MDM policies are applied, the desktop is locked to exam-only mode. The system generates a site readiness report for each location. Any site failing readiness is flagged for manual intervention.
T minus 1 day	Final Verification and Candidate Preparation	Package integrity re-verified at each site (checksum validation). Local server health check at edge sites (disk space, memory, CPU). Candidate lists loaded and verified against registered roll numbers. Attendance sheets generated. Biometric verification devices tested (NADRA connectivity confirmed). Invigilator accounts activated and dashboard access tested. NTC-2 is prepared for switchover to CBT mode the following morning. The system sends automated SMS and email reminders to candidates with their centre, seat number, and reporting time.

4.6.2 Examination Day Operations

Examination day is the most operationally sensitive window. Every action during this phase is time-critical and must function without interruption. The system must handle biometric verification, exam delivery, real-time monitoring, and multi-shift operations, all while maintaining complete security of question content and candidate responses.

Timing	Activity	System Actions
T minus 2 hours	System Activation	NTC-2 switches from EMS/Portal mode to CBT mode. EMS/Portal continues on NTC-1 alone (sufficient for low exam-day portal traffic). NTC-2 and NTC-3 form a load-balanced CBT pair. Edge servers EDGE-1 and EDGE-2 are brought online in exam mode. All SEB terminals at all 8 sites are powered on and confirm connectivity to their designated server.

Timing	Activity	System Actions
T minus 1 hour	Staff Deployment and Biometric Setup	Invigilators log into the invigilator dashboard at their assigned site. Biometric verification devices are powered on and NADRA connectivity is confirmed (for centralized sites, via NTC; for edge sites, via local VPN if available, otherwise offline verification using pre-loaded candidate biometric data). Exam hall seating is verified against the system-generated seating plan.
T minus 30 minutes	Candidate Check-In	Candidates arrive and are verified through biometric authentication (fingerprint or CNIC-based verification via NADRA). The system matches the candidate against their registered profile and photograph. Verified candidates are seated at their assigned terminal. The invigilator dashboard shows real-time check-in progress: how many candidates have arrived, how many are verified, how many are pending. Candidates who fail biometric verification are flagged for manual identity check by the senior invigilator.
T minus 0 (Start)	Exam Commencement	The decryption key is released through a secure channel separate from the package distribution channel. At the scheduled start time, the system decrypts the exam package in server memory (never written to disk in plaintext). Each candidate's SEB terminal receives a randomized question sequence: question order and option order are both randomized so that no two adjacent candidates see the same arrangement. The exam timer starts. Candidates see an intuitive interface with question navigation panel, flag-for-review option, and a countdown timer with warnings at configurable intervals.
During Exam	Live Monitoring	The invigilator dashboard displays real-time status for every candidate: current question, time remaining, questions answered, questions flagged, and anomaly indicators (e.g., unusually fast completion, extended inactivity). For centralized sites, this data streams to NTC-3 in real time. For edge sites, monitoring runs locally on EDGE-1/EDGE-2; if WAN is available, a copy streams to NTC-3 for central oversight. Candidate responses are auto-saved continuously and encrypted on the server. If a candidate's terminal fails mid-exam, the device swap function allows the candidate to move to a replacement terminal and resume from exactly where they left off, with no data loss.
During Exam	Incident Handling	The system logs all incidents: terminal failures, biometric re-verification requests, candidate complaints, time extensions granted by the senior invigilator. Each incident is tagged with the candidate roll number, terminal ID, timestamp, and invigilator action. Time extensions, if granted, are applied to the individual candidate's timer only and recorded in the audit trail.
Exam End	Auto-Submission	When the timer expires, the exam auto-submits for each candidate. Candidates who finish early can submit manually. After submission, the candidate's screen displays a confirmation message and the terminal locks. No candidate can re-enter the exam after submission. All responses are encrypted and stored on the exam server (NTC-3 for centralized sites, EDGE-1/EDGE-2 for edge sites).

Timing	Activity	System Actions
Between Shifts	Multi-Shift Operations	If multiple shifts are scheduled on the same day, the system performs a secure data wipe on all candidate terminals between shifts: all cached exam data, browser history, and temporary files are purged. The next shift's candidate list is loaded. Fresh biometric verification is required for each shift. The exam package for the next shift may use a different question set (assembled from the same QDB pool but with different randomization) to prevent information leakage between shifts.

4.6.3 Post-Examination and Recruitment Completion

Post-examination processing covers everything that happens after the last candidate submits, from result synchronization and compilation through psychometric analysis and official result publication.

Timing	Activity	System Actions
T plus 0 (Same Day)	Result Synchronization	Centralized site results are already on NTC-3. For edge sites (Gilgit, Quetta): the encrypted result files are synced from EDGE-1/EDGE-2 to NTC-3 as soon as the WAN link is confirmed. Each result file is transmitted with a checksum. NTC-3 verifies the checksum on receipt; if it does not match, the sync is retried. Once sync is confirmed successful, the local exam cache on the edge server is securely wiped (cryptographic erasure). The system generates a sync confirmation report showing: site, number of candidates, result file size, checksum match status, and wipe confirmation.
T plus 0 (Same Day)	NTC-2 Switchback	After all exam results are confirmed received, NTC-2 switches back from CBT mode to EMS/Portal mode. NTC-1 and NTC-2 resume normal load-balanced EMS operations.
T plus 1 day	Result Compilation	Raw MCQ scores are compiled automatically from the answer data. The system matches candidate responses against the answer key (which was part of the encrypted exam package). Scores are calculated, and a preliminary merit list is generated. Secrecy Wing reviews the compilation. Anomalies (e.g., identical answer patterns between adjacent candidates, statistically improbable scores) are flagged for investigation. Commission approval is obtained before results are finalized.
T plus 1 to 3 days	Quality Assurance	PISC conducts QA checks on the result data: sample verification of score calculations, audit trail completeness review, and system log analysis for any anomalies during the exam. Any discrepancies identified trigger a re-verification workflow.
T plus 1 to 7 days	Psychometric Analysis	The psychometric analytics engine on SEC-2 processes the exam results against the question bank. For each question used in the exam, the system computes: facility index (how many candidates answered correctly), discrimination index (how well the question differentiates between high and low performers), discrimination efficiency, and reliability coefficient. Questions falling below configured psychometric thresholds are automatically flagged for review or retirement.

Timing	Activity	System Actions
		in the QDB. This analysis feeds back into the question bank to improve future paper generation.
T plus 3 to 7 days	Result Publication	After Commission approval, results are published: merit lists uploaded to the candidate portal and website, individual marksheets made available to candidates through their authenticated portal login, SMS and email notifications sent to all candidates with their result status. The system records the exact publication timestamp and the authorizing officer.
T plus 7 to 30 days	Post-Result Processing	Recounting requests are received and processed through the system. The system retrieves the candidate's stored responses and re-verifies them against the answer key. Recounting results are recorded with a before/after comparison. Grievances and representations are submitted through the candidate portal (as specified in CC-07). Shortlisting for the next phase (document scrutiny, interviews) is triggered based on the merit list cutoff.

4.7 Vendor Deployment Responsibilities

Software Vendor Provides	FPSC / Hardware Vendor / NTC Provides
Application software (all 6 modules, website, portal)	Physical servers, storage, and virtualization platform
Database schema design, creation, and optimization	Server OS installation and licensing
Application deployment on provisioned VMs	VM provisioning on HCI stack
CBT edge server software package for Gilgit and Quetta	Physical edge servers and their OS/network configuration
SEB configuration and lockdown policies	SEB software licenses and desktop MDM enrollment
NADRA API integration (software side)	NADRA VPN tunnel and network connectivity
Data migration scripts and validation	Access to existing Oracle 11g/MySQL databases
User training (FPSC staff, CBT site staff)	Training venue, trainee availability, logistics
Technical documentation and manuals	N/A
Bug fixes and patches during 24-month warranty	Hosting, bandwidth, and infrastructure maintenance
Source code repository with full commit history	N/A

4.8 Software Component to Server Mapping

The following table shows which categories of software components run on each server. Specific product choices (e.g., PostgreSQL vs MySQL, Nginx vs Apache, Redis vs Memcached) are at the Vendor's discretion subject to the constraints in Section 6.1. This table defines where each category of component must be deployed.

4.8.1 Per-Server Software Components

Server	Role	Software Components	Description
NTC-1	EMS + Portal (Active)	Application Server Web Server / Reverse Proxy Notification Service Template Engine Cache (shared)	Hosts the application runtime for GR, CE, UEM, and Supporting modules along with the candidate portal and public website. Runs the web server that handles incoming requests and the notification service for SMS/email dispatch. Shares a distributed cache with NTC-2 for session management.
NTC-2	EMS + Portal / CBT Swing	Application Server Web Server / Reverse Proxy CBT Application (exam-day) Cache (shared)	Mirrors NTC-1 during normal operations for load balancing. On exam day, the EMS application is stopped and the CBT exam delivery application is started. Session state is stored in the external cache so candidates are not affected by the switchover.
NTC-3	CBT Engine + Reporting	CBT Application Server Reporting/Analytics Engine Message Queue Consumer Cache (shared)	Runs the CBT exam delivery engine full-time. Manages exam sessions, question delivery, answer collection, and the invigilator dashboard. After exams, runs result compilation and reporting jobs. Consumes async messages (result sync from edge nodes, report generation requests) from the message queue.
NTC-4	Primary Database	RDBMS (EMS transactional database) RDBMS (Reporting replica) Message Queue Broker Backup Agent Database Monitoring	Hosts the primary transactional database for all EMS modules and the candidate portal. A read replica on the same server handles reporting queries without impacting transactions. The message queue broker routes asynchronous tasks across all Tier 1 servers. Backup agent runs automated daily backups with integrity verification. Replicates asynchronously to DR-3.
SEC-1	QDBMS Primary	QDBMS Application QDBMS Database (encrypted) Exam Package Builder NADRA Integration Gateway	Runs the Question Data Bank application and its encrypted database on the same server for physical isolation. The exam package builder assembles questions into encrypted CBT packages. The NADRA gateway provides a secure VPN endpoint for

Server	Role	Software Components	Description
			CNIC and biometric verification. No public internet exposure.
SEC-2	QDBMS Standby	QDBMS Application (hot standby) QDBMS Database (read replica) Psychometric Analytics Engine Backup Agent	Hot standby mirror of SEC-1 that takes over if SEC-1 fails. During normal operations, serves read-only psychometric reports and question bank statistics. Sends nightly encrypted backups to DR-4.
EDGE-1 (Gilgit)	CBT Edge Node	CBT Application (local mode) Local Database (exam cache) SEB Management Result Sync Agent	Runs the same CBT application as NTC-3 but in local/edge mode. Stores the encrypted exam package locally before exam day. Decrypts and serves exams to 50 SEB terminals. After the exam, encrypts results and syncs to NTC-3 when WAN is available. Local exam cache is wiped after successful sync.
EDGE-2 (Quetta)	CBT Edge Node	CBT Application (local mode) Local Database (exam cache) SEB Management Result Sync Agent	Same configuration as EDGE-1 but sized for 100 concurrent candidates.
DR-1	EMS + Portal (Warm Standby)	Application Server (standby) Web Server (standby)	Receives asynchronous state replication from NTC-1/NTC-2. Can be activated within the agreed RTO to take over EMS and portal operations.
DR-2	CBT Engine (Standby)	CBT Application (standby) Reporting Engine (standby)	Receives replication from NTC-3. Can take over exam orchestration if NTC Islamabad is entirely unavailable.
DR-3	Database (DR Replica)	RDBMS (async replica) Backup Agent Database Monitoring	Asynchronous replica of NTC-4. Connected to SAN storage at NTC Lahore.
DR-4	QDB Backup + Spare	Encrypted Backup Storage Spare Capacity	Receives nightly encrypted QDB snapshots from SEC-2. Does not run an active QDBMS instance. Available as a spare if any other DR server fails.

4.8.2 Cross-Server Services

The following services span multiple servers and are not confined to a single node:

Service	Location	Purpose
Distributed Cache / Session Store	NTC-1, NTC-2, NTC-3	Stores user sessions and frequently accessed data externally so any application server can serve any request. Enables the exam-day switchover of NTC-2 without session loss.
Message Queue	Broker on NTC-4, consumers on all Tier 1 servers	Routes asynchronous tasks: notification dispatch, report generation, result sync from edge nodes, psychometric analysis triggers. Prevents these operations from blocking user-facing requests.
Database Replication	NTC-4 to DR-3 (async), SEC-1 to SEC-2 (sync), SEC-2 to DR-4 (nightly backup)	Ensures data redundancy across sites. Transactional data replicates to DR continuously. QDB data replicates within Tier 2 in real time and to DR as encrypted nightly snapshots.
SEB (Safe Exam Browser)	Candidate desktops at all 8 CBT sites	Lockdown browser installed on candidate terminals. Managed centrally through MDM. Connects to NTC-3 (centralized sites) or EDGE-1/EDGE-2 (edge sites) during exams.
MDM (Mobile Device Management)	Management console on NTC-1, agents on all candidate desktops	Controls and secures candidate desktops at CBT labs. Enforces SEB lockdown, remote wipe capability, policy-based application control.
Monitoring and Logging	Agents on all servers, central dashboard on NTC-1	Collects system health metrics, application logs, and security events from all servers across all tiers. Central dashboard for FPSC IT staff and vendor support.

Note: The Vendor's technical proposal shall include a detailed software architecture diagram mapping specific technology choices to this server layout. The architecture shall comply with the deployment model described in Sections 4.2 through 4.5 and the technology constraints in Section 6.1.

4.9 Examination Data Lifecycle

This section traces the lifecycle of examination data and candidate data through the system, from creation to long-term retention. It specifies where data resides at each stage, what state it is in, who can access it, and how long it is kept.

4.9.1 Question and Examination Data

The following table tracks question data from the moment it is authored in the Question Data Bank through exam delivery, result compilation, and long-term retention.

Stage	Location (Server)	Data State	Who Can Access	Retention
Authoring	SEC-1 (QDBMS)	Draft. Encrypted at rest (AES-256). Stored in QDB.	Question author only. No other role can see the question at this stage.	Retained indefinitely in QDB. Never deleted; retired questions remain in the bank for audit purposes.
Review and Approval	SEC-1 (QDBMS)	Under Review, then Approved. Still encrypted. Not yet available for paper generation.	Reviewer sees assigned questions only. Approver sees assigned questions only. No single role sees a complete set.	Status changes logged with user, timestamp, and action.
Paper Generation	SEC-1 (QDBMS), network-isolated mode	Active. Questions assembled into an exam paper. SEC-1 disconnects from all networks during assembly. Dual authorization required.	Two authorized role holders must approve simultaneously. No one else has access during this window.	Paper composition (which questions were selected) logged permanently. The assembled paper is encrypted immediately.
Package Distribution (T minus 7 days)	SEC-1 to NTC-3 and EDGE-1/EDGE-2	Encrypted exam package (AES-256). Decryption key held separately and not transmitted with the package.	System administrators at CBT sites can verify package integrity (checksum) but cannot read the contents.	Package receipt and integrity verification logged at each site.
Exam Delivery (T minus 0)	NTC-3 (centralized sites), EDGE-1/EDGE-2 (edge sites)	Decrypted in memory at exam time. Questions served to SEB terminals one at a time. Never stored in plaintext on disk.	Candidates see only their assigned questions through SEB. Invigilators see progress and status but not	Every question served and every answer submitted logged with timestamp.

Stage	Location (Server)	Data State	Who Can Access	Retention
			question content.	
Answer Collection (during exam)	NTC-3 or EDGE-1/EDGE-2	Candidate responses stored encrypted on the exam server. Auto-saved continuously. Auto-submitted on timer expiry.	No one can access individual responses during the exam. Invigilator dashboard shows progress only.	Each save event logged.
Post-Exam Sync (T plus 0)	EDGE sites sync to NTC-3. Centralized site data already at NTC.	Results encrypted at source. Transmitted with integrity checksum. Local exam cache on edge servers securely wiped after successful sync.	System-level process only. No human access to raw answers at this stage.	Sync timestamp, checksum match, and cache wipe confirmation logged.
Result Compilation (T plus 1 day)	NTC-3 and NTC-4	Raw scores compiled into merit lists. Stored in EMS transactional database on NTC-4.	Secrecy Wing for compilation. Commission for approval. IT Wing for publication. Role-based access enforced.	Compilation actions logged. Commission approval recorded with timestamp and approving authority.
Psychometric Analysis	SEC-2 (read replica)	Question performance statistics (facility index, discrimination index, reliability) computed from exam results. Fed back into QDB as metadata on each question.	QDB administrators and authorized CR and C Wing staff.	Analysis results stored permanently against each question in QDB.
Published Results	NTC-1 (Portal), NTC-4 (Database)	Merit lists, marksheets, and result notifications published on candidate portal and website.	Public (merit lists). Candidates (individual marksheets via authenticated portal). FPSC staff (detailed reports via EMS).	Published results retained permanently as part of FPSC's official record.

4.9.2 Candidate Data

The following table tracks candidate information from initial application through final nomination and beyond.

Stage	Location (Server)	Data Created	Who Can Access	Retention
Application and Registration	NTC-4 (EMS Database)	Candidate profile, personal data, education, experience, uploaded documents, photograph, fee payment record.	Candidate (own data via portal). Relevant wing staff (role-based). IT Wing (administration).	Application data retained permanently. Candidate photographs retained for 2 years from the date of the relevant examination, then archived or purged per FPSC policy.
Admission and Scheduling	NTC-4 and NTC-1 (Portal)	Roll number, centre allocation, admit card, attendance sheet data.	Candidate (admit card via portal). Relevant wing and IT Wing.	Retained permanently as part of examination record.
Identity Verification (NADRA)	SEC-1 (NADRA Gateway)	Biometric verification result (match/no-match). Raw biometric data is not stored; only the verification outcome is recorded.	System-level automated check. Result visible to relevant wing staff.	Verification outcome retained permanently. No biometric templates stored in FPSC systems.
Document Scrutiny	NTC-4 (EMS Database)	Scrutiny results, shortlisting/rejection decision, officer review notes, CV scoring output.	Relevant wing scrutiny officers. Commission (for approval). Candidate (outcome via portal).	Retained permanently. Audit trail of every scrutiny decision preserved.
Interview and Personal Hearing	NTC-4 (EMS Database)	Interview scores, panel composition, personal hearing decisions, viva voce records.	Program Management (scheduling). Commission (conduct). Relevant wing (recording). Candidate (outcome via portal).	Retained permanently.
Final Nomination	NTC-4 (EMS Database), NTC-1 (Portal)	Nomination letters, joiner/non-joiner tracking, alternate nominations.	FS Directorate. Commission. Requisitioning department (intimation).	Retained permanently.

Stage	Location (Server)	Data Created	Who Can Access	Retention
			Candidate (notification).	
Grievances and Appeals	NTC-4 (EMS Database)	Grievance submissions, tracking status, hearing records, decisions.	Candidate (own grievance status). Relevant wing. Commission.	Retained permanently.

4.9.3 Data Retention Summary

The table below consolidates retention periods for all major data categories. These periods are based on current FPSC practice and government regulations as stated in the filled questionnaire. FPSC may revise these periods during the BPR phase or through a separate data retention policy.

Data Category	Retention Period	Basis
Candidate application data (text fields)	Permanent	Official record of every application received by FPSC.
Candidate photographs	2 years from date of relevant examination	Current FPSC policy due to storage constraints. To be reviewed after new system deployment.
Examination answer responses (CBT)	Permanent	Required for recounting requests, appeals, and audit.
Question bank items (QDB)	Permanent (never deleted)	Retired questions remain for audit, psychometric history, and reuse prevention.
Exam packages (encrypted)	Purged from edge servers after sync. Archived on NTC-3 for 1 year.	Edge cache is wiped immediately after successful result sync. Central archive retained for dispute resolution.
Merit lists and result records	Permanent	Official FPSC record. Required for Annual Report and historical reference.
Audit logs (all modules)	Permanent	Tamper-evident logs required for accountability and compliance.
Grievance and appeal records	Permanent	Legal and administrative requirement.
Hard copy files (physical)	5 to 7 years (existing FPSC practice)	Physical archival per government regulations. Not managed by this software.
System backups (daily)	7 days point-in-time recovery (EMS). Nightly snapshots (QDB) retained per FPSC backup rotation policy.	As specified in BR-04 and BR-06.

The system shall support configurable retention rules that allow FPSC administrators to set per-category retention periods and automated archival or purge actions without code changes. The Vendor shall implement retention controls during development and configure them based on the retention periods confirmed by FPSC during BPR.

5. Non-Functional Requirements

This section specifies measurable non-functional requirements that apply across the entire system. The Vendor shall address each requirement in the Compliance Matrix (Annexure A) and demonstrate compliance during UAT.

5.1 Performance

Ref	Metric	Target and Measurement
P-01	Portal page load time	Less than 2 seconds (95th percentile). Automated load test simulating 25,000 concurrent portal users.
P-02	CBT question rendering	Less than 1 second. Measured at SEB client terminal under exam conditions.
P-03	API response time	Less than 500 ms (95th percentile). Load test with simulated concurrent API traffic.
P-04	Concurrent CBT capacity	1,250 simultaneous test-takers (load-tested for 3,000). Full-capacity exam simulation across all sites.
P-05	Concurrent portal capacity	25,000 simultaneous users. Peak registration/result-day traffic simulation.
P-06	Result compilation (MCQ)	Same day, within 4 hours of last exam session closing.
P-07	Question paper generation	Less than 1 hour from dual-authorization approval to encrypted package availability.
P-08	Search response time	Less than 1 second for candidate/requisition/question searches with production-representative data volume.
P-09	Report generation	Less than 30 seconds for standard reports; less than 5 minutes for complex ad-hoc reports. Measured against production-scale dataset.
P-10	Notification delivery	SMS/Email dispatched within 60 seconds of trigger event. Measured from workflow event to gateway submission.

5.2 Availability and Reliability

Ref	Requirement
A-01	Overall system availability: 99.99% uptime measured monthly, excluding pre-agreed scheduled maintenance windows.
A-02	Scheduled maintenance window: maximum 4 hours per month, between 00:00 and 04:00 PST, with 72 hours advance notice to FPSC.
A-03	CBT exam window availability: zero tolerance for system-initiated downtime during a scheduled exam window. Any exam-day failure attributable to software is a Critical defect.
A-04	Failover, Tier 1 (NTC-2/NTC-3): if either CBT server fails during an exam, the surviving server shall assume full load within 2 minutes without candidate session loss.

Ref	Requirement
A-05	Failover, Tier 3 (Edge): edge servers shall operate independently of WAN. No candidate impact from WAN disconnection during exam.
A-06	Failover, Tier 2 (QDB): if SEC-1 fails, SEC-2 shall assume QDBMS operations within 5 minutes.

Note: DR activation and RPO/RTO requirements are specified in BR-06 (Section 5.5).

5.3 Security

The system handles sensitive examination data including question papers, candidate personal information, and examination results. The following security requirements are mandatory, derived from PC-1 security specifications and ISO 27001 aligned controls.

The table below sets out, for each security area, the business objective and the corresponding requirement reference. The mandatory security requirements follow.

Security Area	Business Objective	How Current RFP Addresses It
Access Control & User Security	Prevent unauthorized access to sensitive examination functions	Role-based access controls and workflow permissions defined in Sections 3.6, 3.8, and 4.9
Auditability & Accountability	Ensure traceability of all examination and approval activities	Audit trail and workflow logging mechanisms included in CC-01 and Section 4.8
Examination Confidentiality	Protect question papers and examination data from leakage or tampering	Encrypted QDBMS, secure CBT delivery, and controlled paper workflows in Sections 3.6 and 4.6
Secure Examination Operations	Maintain integrity of CBT operations and candidate sessions	Secure CBT environment and controlled synchronization architecture in Section 3.6
Data Protection	Protect candidate, examination, and operational records	Centralized data lifecycle and secure storage architecture in Sections 4.7 and 4.8
Disaster Recovery & Continuity	Ensure continuity of examination operations during failures	DR and backup architecture defined in Section 4.7
Workflow Governance	Ensure secure approvals and controlled operational workflows	Workflow controls and approval mechanisms included in Sections 3.8 and 4.9
Secure External Connectivity	Protect integrations with NADRA, HEC, and external platforms	External integration controls covered in Section 3.9
Operational Monitoring	Enable centralized visibility and oversight	Dashboards and monitoring services included in SUP-6.6 and TAB modules

Table: Security Coverage Matrix

Ref	Requirement
S-01	Encryption at rest: all data encrypted using AES-256. QDB content must use field-level encryption in addition to disk-level encryption.
S-02	Encryption in transit: TLS 1.3 minimum for all communications. No unencrypted data transmission between any system components.
S-03	Authentication: multi-factor authentication (MFA) mandatory for all administrative, QDB, and internal staff access. Candidate portal authentication via CNIC + password + OTP.
S-04	Authorization: role-based access control (RBAC) with principle of least privilege. Roles and permissions configurable by administrator without code changes across all modules.
S-05	Session management: configurable session timeout. Secure cookies (HttpOnly, Secure, SameSite). Anti-CSRF tokens on all state-changing operations.
S-06	Password security: passwords hashed using bcrypt or Argon2. Password complexity policy configurable by administrator. Account lockout after configurable failed attempts.
S-07	OWASP compliance: application must be hardened against OWASP Top 10 vulnerabilities: SQL injection, XSS, CSRF, broken authentication, sensitive data exposure, XML external entities, broken access control, security misconfiguration, insecure deserialization, and insufficient logging.
S-08	Input validation: all user inputs validated on both client and server side. File upload restricted to allowed types with content-type verification and malware scanning.
S-09	API security: API authentication via tokens. Rate limiting. Input validation on all endpoints. API keys rotatable without downtime.
S-10	Security event logging: all security-relevant events logged (login/logout, failed authentication, privilege changes, data access, configuration changes). Logs retained per FPSC data retention policy. (Business data audit trail specified separately in CC-01.)
S-11	Vulnerability management: Vendor shall conduct monthly automated vulnerability scans. Remediation SLA: Critical, 24 hours; High, 7 days; Medium, 30 days.
S-12	Penetration testing: Vendor shall facilitate a penetration test by an independent third party at least once before go-live and annually during warranty. Scope and frequency confirmed by FPSC.
S-13	Secure development: Vendor shall follow SSDLC: secure coding guidelines, mandatory code review for security-sensitive modules, dependency vulnerability scanning, and no deployment of components with known critical vulnerabilities.
S-14	HTTPS enforcement: all web interfaces served over HTTPS only. HTTP requests redirected to HTTPS. HSTS headers enabled.
S-15	QDB-specific controls: split-knowledge access (no single role views a complete paper). Dual authorisation for paper generation and question export. Paper generation engine operates in network-isolated mode. All QDB access logged with user, timestamp, question ID, and action.

5.4 Scalability

Scalability requirements are specified in Section 6.4 (Software Architecture): SA-01 (horizontal scaling), SA-02 (edge node configurability), SA-08 (configuration-driven behavior). QDB capacity (1,000,000+ items) is specified in QDB-5.1.

5.5 Backup and Recovery

Ref	Requirement
BR-01	The application shall support automated daily database backups without requiring downtime or manual intervention.
BR-02	Backup integrity shall be automatically verified after each backup cycle (checksum validation or test restore).
BR-03	The Vendor shall provide and document backup/restore procedures. FPSC IT staff must be able to perform a full restore using the documented procedure.
BR-04	The system shall support point-in-time recovery for the EMS database to any point within the last 7 days.
BR-05	Annual DR drill: the Vendor shall participate in a full system restore test at the NTC Lahore DR site once per year during the warranty period.
BR-06	Disaster Recovery: the system shall meet a Recovery Point Objective (RPO) of 15 minutes or less and a Recovery Time Objective (RTO) of 4 hours or less for the production EMS database at the NTC Lahore DR site. During an active CBT examination window, CBT edge nodes shall continue to operate independently of the central site (see A-05), so that a central-site failure does not interrupt an examination in progress. The Vendor's software architecture must support asynchronous database replication to the DR site and a documented, tested failover procedure.

5.6 Data Migration

Ref	Requirement
DM-01	Migrate all historical data from the existing Oracle 11g/MySQL-based Online Recruitment System into the new platform. This includes approximately 17 years of application data (approximately 100 GB).
DM-02	Migrate post-examination data currently maintained in Excel/Word files across individual wings/directorates, where such data is available in structured or semi-structured format.
DM-03	Provide a documented data migration plan including: source-to-target field mapping, transformation rules, validation criteria, rollback strategy, and migration schedule.
DM-04	Execute migration in a staged approach: test migration on sample dataset, full migration on staging environment with validation, production migration with FPSC sign-off.
DM-05	Post-migration reconciliation: the Vendor shall produce a reconciliation report proving record counts, data integrity, and completeness match between source and target. FPSC shall sign off before the old system is decommissioned.

5.7 Accessibility and Compatibility

Ref	Requirement
AC-01	Content accessibility: the candidate-facing portal and website shall comply with WCAG 2.1 Level AA, including provisions for persons with special needs.
AC-02	CBT interface shall provide high-contrast mode and configurable font size for candidates with visual impairments.
AC-03	The candidate portal and website shall be mobile-responsive, functioning correctly on screen sizes from 360px width upward.
AC-	Browser compatibility: the system shall support the latest two major versions of Chrome, Firefox, Edge, and Safari. The CBT interface must be compatible with the SEB rendering

Ref	Requirement
04	engine.
AC-05	Localization readiness: the initial deployment language is English. The system architecture shall support the future addition of Urdu and other languages through configuration only, without code changes, including support for right-to-left text rendering where applicable.

6. Technical Requirements

6.1 Technology Stack

The use of open-source technologies is preferred, in line with the approved project documentation, to reduce licensing costs and promote long-term sustainability. Licensed components are permitted where necessary. The Vendor's choice of technology stack is at their discretion, subject to the following constraints:

Ref	Requirement
TS-01	All proposed technologies must be actively maintained with a demonstrable support roadmap of at least 5 years from proposal submission.
TS-02	No proprietary hardware dependency. The system must be deployable on standard x86-64 Linux servers. No technology requiring specific hardware from a single manufacturer.
TS-03	Database: enterprise-grade RDBMS with ACID compliance, high-availability support, encryption at rest, and replication capability. PostgreSQL, MySQL/MariaDB, or Microsoft SQL Server are acceptable. The Vendor shall justify the choice.
TS-04	If based on an open-source framework, the Vendor must demonstrate at least one successful deployment at institutional scale (10,000+ registered users) with verifiable client reference.
TS-05	All third-party libraries, frameworks, and components must have active security patch support. No end-of-life or unmaintained dependencies in production.
TS-06	If licensed software components are proposed, the Vendor shall itemize all license costs in the Financial Proposal (Annexure B) with license type, duration, and renewal terms.
TS-07	The system must run on Linux-based server operating systems. Windows Server may be used for specific components if justified, but the core stack shall not require Windows.

6.2 Source Code and Intellectual Property

Ref	Requirement
IP-01	All source code, documentation, designs, databases, algorithms, configurations, and deliverables produced under this contract shall be the sole and exclusive intellectual property of FPSC, Government of Pakistan. The Vendor shall execute an IP assignment agreement at contract signing.
IP-02	Source code shall be delivered in a Git repository with full commit history, branching structure, and release tags intact. Handed over at each milestone and in final form at project completion.
IP-03	The Vendor shall not retain copies of any project deliverables, source code, or FPSC data after project completion, except as required by law.
IP-04	The Vendor shall not use any project-related information for any purpose other than this project, nor share, publish, or disclose any project information without prior written consent from FPSC.
IP-05	Where open-source components are incorporated, the Vendor shall provide a complete list of all libraries used, their licenses, and confirmation that all licenses are compatible with FPSC's exclusive ownership of the custom-developed code.

6.3 Development Standards

Ref	Requirement
DS-01	Version control: Git-based repository with documented branching strategy (e.g., GitFlow or trunk-based). All development committed to the repository; no code outside version control.
DS-02	Code review: all code changes reviewed by at least one senior developer before merge. Code review records available in repository history.
DS-03	Testing: minimum 80% unit test coverage for business logic. Integration tests for all API endpoints. End-to-end tests for all critical user workflows (candidate registration, exam delivery, result compilation, paper generation).
DS-04	CI/CD: automated build, test, and deployment pipeline. Every commit to main branch triggers automated tests. Deployment to staging and production repeatable and scripted; no manual deployment steps.
DS-05	Code documentation: inline comments for complex logic, README files for each module/service. Stale documentation is treated as a defect. (API documentation requirements specified in SA-03.)
DS-06	Technical documentation: system architecture document, database schema with ER diagrams, API documentation, deployment guide (fresh deployment and upgrade), and system administrator manual. Updated at each milestone.
DS-07	Dependency management: all third-party dependencies managed through a package manager (e.g., npm, pip, Maven). Dependencies pinned to specific versions. No manual file inclusion of third-party code.

6.4 Software Architecture

The following architecture requirements are authoritative. Where the same concept appears elsewhere in this document, this section prevails.

Ref	Requirement
SA-01	The application tier shall be stateless and horizontally scalable. Session state managed externally (database or distributed cache) to enable any node to serve any request and support the exam-day workload switchover described in Section 4.2. New nodes added without architectural changes or downtime.
SA-02	The CBT component shall support both centralized mode (terminals connect to NTC) and edge mode (terminals connect to local server) from the same codebase. Switching between modes shall require configuration changes only. New CBT sites onboarded by deploying an edge node package or configuring centralized access; no code changes required.
SA-03	The system shall expose RESTful APIs for all core functions with auto-generated OpenAPI/Swagger documentation. APIs shall be versioned to support backward compatibility during upgrades.
SA-04	Modular design: individual modules (GR, CE, UEM, CBT, QDBMS, Website, Supporting) shall be independently deployable and updatable. An update to one module shall not require redeployment of unrelated modules.
SA-05	The system shall use a message queue or event-driven architecture for asynchronous operations (notifications, report generation, result compilation, data sync from edge nodes) to prevent blocking of user-facing operations.
SA-06	Database design shall follow normalization best practices with appropriate indexing. The Vendor shall produce a complete entity-relationship diagram and data dictionary as part of design deliverables. Database shall support read replicas for reporting and analytics without impacting transactional performance.

Ref	Requirement
SA-07	The architecture shall cleanly separate the presentation layer, business logic layer, and data access layer. Business rules shall not be embedded in database stored procedures or frontend code.
SA-08	Configuration-driven behavior: workflow rules, approval chains, notification triggers, classification taxonomies, report templates, examination types, and user roles shall all be administrator-configurable without code deployment. New examination types shall be configurable through the UEM admin interface without code changes.

7. Project Milestones and Deliverables

7.1 Delivery Phases

The software development is to be completed within 12 months from contract signing, followed by a 24-month warranty and operations support period. The Vendor shall propose a detailed project plan in their technical proposal; the phasing below represents the minimum expected structure.

Phase	Title	Timeline	Key Deliverables
1	Inception, BPR and Design	Months 1 to 3	Inception report with detailed project plan and WBS. Business Process Re-engineering (BPR): engage all FPSC wings/directorates, map existing workflows, identify redundancies, propose optimized processes. Software Requirements Specification (SRS) based on BPR findings, subject to FPSC approval through PISC review. System architecture and design documents. Database design with entity-relationship diagrams and data dictionary. UI/UX wireframes and prototypes for key interfaces (portal, CBT, QDB, dashboards). Security architecture document. Test plan and test strategy. Data migration plan.
2	Core EMS Development	Months 3 to 7	Module 1 (GR): requisition through final nomination workflow. Module 2 (CE): CSS lifecycle including MPT, written exam, psychological assessment, viva voce, group allocation. Module 3 (UEM): configurable exam type engine. Module 6 (Supporting): HR duty deployment, inventory, transport, attendance/leave, decision support dashboards. Cross-cutting: RBAC, audit trail, workflow engine, notification engine, reporting engine, template engine. Unit tests and integration tests for all developed modules.
3	CBT and QDBMS Development	Months 4 to 8	Module 5A (QDBMS): question authoring, classification, versioning, workflow, dual-authorization, paper generation engine, psychometric analytics, encryption. Module 5B (CBT): exam delivery engine, SEB integration, randomization, timer, biometric integration, invigilator dashboard, offline/edge mode, exam package encryption, device swap, multi-shift. Edge node software package for Gilgit and Quetta deployment. Unit tests and integration tests.
4	Portal, Website and Integration	Months 5 to 9	Module 4 (Website and Candidate Portal): CMS, public content, candidate self-service, mobile-responsive design. NADRA integration (biometric verification). Payment gateway integration (online fee collection). SMS/Email notification gateway integration. Data migration: test migration on sample dataset, validation. End-to-end integration testing across all modules.
5	Pilot Deployment	Months 9 to 10	Deployment at Islamabad CBT site (250 seats) and FPSC HQ. Pilot exam with real or simulated candidates. Performance tuning based on pilot results. Security audit and penetration test (pre-go-live). Data migration: full migration on staging environment with FPSC validation. Bug fixes from pilot feedback.

Phase	Title	Timeline	Key Deliverables
6	Full Deployment and UAT	Months 10 to 11	Rollout to remaining 7 CBT sites (including edge servers at Gilgit and Quetta). Load testing at full scale (3,000 concurrent CBT users). DR site setup and failover test at NTC Lahore. User Acceptance Testing (UAT) by FPSC end-users facilitated by PISC. Production data migration with reconciliation report and FPSC sign-off.
7	Go-Live and Stabilization	Month 12	Production go-live. 30-day stabilization period with on-site vendor support. Priority bug fixes and performance optimization. Training delivery (see Section 7.3). Final source code handover with complete documentation. Project closure report.
8	Warranty and O and M	Months 13 to 36	24-month warranty support commencing from go-live acceptance. Bug fixes, security patches, and minor enhancements at no additional cost. Quarterly health checks and performance reviews. Annual DR drill participation. Annual penetration test facilitation. Knowledge transfer to FPSC IT staff throughout the warranty period.

Phases 2, 3, and 4 overlap by design. The Vendor is expected to run parallel development workstreams with separate teams for EMS, CBT/QDBMS, and Portal/Integration. The Vendor's technical proposal shall include a Gantt chart demonstrating how parallel workstreams are staffed and managed.

7.2 Deliverable Documents

The following documents shall be produced by the Vendor and submitted in both soft copy (editable Word/PDF) and hard copy (one set) at the indicated milestone:

Document	Delivery Phase	Approval By
Inception Report (project plan, WBS, risk register)	Phase 1	FPSC via PISC review
BPR Report (current-state mapping, gap analysis, proposed workflows)	Phase 1	FPSC via PISC review
Software Requirements Specification (SRS)	Phase 1	FPSC via PISC review
System Architecture Document	Phase 1, updated each phase	PISC review
Database Design (ERD, data dictionary, schema)	Phase 1, updated each phase	PISC review
UI/UX Wireframes and Prototypes	Phase 1	PISC ->FPSC user feedback
Security Architecture Document	Phase 1	PISC review
Test Plan and Test Strategy	Phase 1	PISC review
Data Migration Plan	Phase 1	PISC review
Monthly Progress Reports	Every month, by 5th	PISC review -> PD / PMC

Document	Delivery Phase	Approval By
Test Reports (unit, integration, system, UAT)	Phases 2 to 6	PISC review
Security Audit / Penetration Test Report	Phase 5	PISC review
Data Migration Reconciliation Report	Phase 6	PISC Review -> FPSC sign-off
User Manuals (admin, staff, CBT site, candidate)	Phase 7	FPSC via PISC review
Training Materials (print, digital, video)	Phase 7	FPSC via PISC review
Source Code Repository (Git, full history)	Each milestone + Phase 7	PISC verification
Technical Documentation Package (architecture, API, deployment, admin)	Phase 7 (final)	PISC review
Project Closure Report	Phase 7	FPSC via PISC review

7.3 Training

The Vendor shall deliver training to FPSC personnel before go-live. Training materials shall be provided in both print and digital formats. Video tutorials shall be produced for key workflows.

Audience	Coverage	Format
FPSC administrative staff	EMS operations: requisition processing, advertisement, scheduling, scrutiny, nominations, correspondence, dashboards, reporting.	Hands-on + classroom
FPSC IT staff	System administration, monitoring, backup/restore, user management, configuration changes, troubleshooting, deployment procedures.	Hands-on + technical lab
CBT site staff	Exam-day operations: SEB management, biometric verification, invigilator dashboard, device swap, incident handling, edge server operations (Gilgit/Quetta).	On-site practical
QDB users (question authors, reviewers, approvers)	Question authoring, review workflow, approval process, psychometric report interpretation, paper generation authorisation.	Hands-on + classroom

The Vendor shall propose a detailed training plan in the technical proposal, including: training duration (hours) per audience group, number of sessions, batch sizes, trainer qualifications, assessment methodology to verify trainee competence, and a schedule aligned with the go-live timeline. The training plan will be evaluated under Section 9 (Technical Evaluation).

7.4 Acceptance Criteria and Process

Each phase deliverable shall follow this acceptance process:

- The Vendor submits the deliverable in draft form to FPSC through the PMC.
- PISC reviews the deliverable against the SRS (or this RFP for Phase 1 deliverables) and provides a technical validation report within 10 working days.
- If PISC identifies deficiencies, the Vendor addresses them and resubmits. The review-resubmit cycle repeats until PISC recommends acceptance.
- Upon PISC recommendation, FPSC issues a formal Milestone Acceptance Certificate.
- Payment for the milestone is released only after issuance of the Acceptance Certificate.
- FPSC reserves the right to reject deliverables that do not meet specified requirements. The Vendor shall rework and resubmit at no additional cost.

Go-Live Acceptance: The system shall be considered accepted for go-live when: (a) all UAT test cases pass, (b) the security audit reports no Critical or High severity findings, (c) data migration reconciliation is signed off by FPSC, (d) load testing confirms the system meets all performance targets in Section 5.1, and (e) training has been delivered to all specified audiences.

Warranty Commencement: The 24-month warranty period begins on the date of the Go-Live Acceptance Certificate. This is distinct from the development contract period.

7.5 Post Go-Live Operations and Maintenance Support

The Vendor shall provide a 24-month operations and maintenance (O and M) support period commencing from the date of the Go-Live Acceptance Certificate. This period is distinct from the development contract and covers the following obligations:

7.5.1 Scope of O and M Support

- Corrective maintenance: fix all software defects, bugs, and errors reported during the O and M period at no additional cost to FPSC. This includes defects in all six modules, the website, the candidate portal, the CBT system, and the QDBMS.
- Adaptive maintenance: update the system to remain compatible with changes in the operating environment, including operating system patches, database engine updates, browser version changes, and third-party API changes (NADRA, payment gateway, SMS gateway).
- Security maintenance: deliver security patches for all identified vulnerabilities. Critical vulnerabilities (CVSS 9.0 and above) shall be patched within 24 hours. Quarterly dependency scanning and update cycle for all third-party libraries.
- Preventive maintenance: quarterly system health checks covering database performance, storage utilization, application logs, security event analysis, and infrastructure utilization trends. Health check reports submitted to PMC within 5 working days.

- Minor enhancements: small functional improvements and configuration changes requested by FPSC that do not exceed 5 person-days of effort per request and 40 person-days cumulative per year. Enhancements exceeding these thresholds shall be quoted separately.

7.5.2 Service Level Agreements

Severity	Definition	Response and Resolution	Support Mode
Critical	System down, data loss, security breach, exam-day failure.	Response within 2 hours. Resolution within 8 hours.	On-site or remote, 24/7.
High	Major functionality impaired, no workaround available.	Response within 4 hours. Resolution within 24 hours.	On-site or remote, business hours extended to 20:00 PST.
Medium	Functionality partially impaired, workaround available.	Response within 1 business day. Resolution within 5 business days.	Remote, business hours.
Low	Cosmetic issues, minor usability improvements.	Response within 3 business days. Resolution in next scheduled release.	Remote, business hours.

7.5.3 Staffing and Availability

- The Vendor shall assign a dedicated support team for the O and M period. At minimum: one support lead (single point of contact for FPSC), two application support engineers, and one database administrator, all available during FPSC business hours (09:00 to 17:00 PST, Monday to Friday).
- During scheduled examination windows (as notified by FPSC at least 14 days in advance), the Vendor shall provide extended support coverage from 06:00 to 22:00 PST, including weekends, with on-site presence at FPSC HQ and at least one CBT site designated by FPSC.
- A 24/7 emergency hotline shall be available for Critical severity issues throughout the O and M period.

7.5.4 Additional O and M Obligations

- Annual disaster recovery drill: the Vendor shall participate in a full system restore test at the NTC Lahore DR site once per year during the O and M period.
- Annual penetration test: the Vendor shall facilitate penetration testing by an independent third party engaged by FPSC. All findings shall be remediated per the vulnerability management SLA in S-11.
- Knowledge transfer: the Vendor shall conduct structured knowledge transfer to FPSC IT staff throughout the O and M period. By the end of month 24, FPSC IT staff shall be able to independently perform: system monitoring, user management, configuration changes, backup and restore, and first-level troubleshooting. The Vendor shall certify FPSC IT readiness before the O and M period ends.

- Transition support: in the final 3 months of the O and M period, the Vendor shall prepare a transition plan and provide handover support to FPSC's internal team or a successor maintenance vendor, if applicable.

8. Preliminary Scrutiny

This section defines the mandatory qualification criteria that every bidder must meet. These are evaluated on a Pass/Fail basis. Failure to meet any single criterion results in disqualification; the bidder's technical and financial proposals will not be evaluated further.

8.1 Mandatory Eligibility Criteria

Ref	Criterion	Evidence Required	Evaluation
E-01	Must be State-owned entity such as professional, autonomous or semi-autonomous organizations or bodies of the Federal or Provincial Governments	SECP certificate/Authorization of provision of such services. NTN/FTN certificate.	Pass / Fail
E-02	Minimum 5 years of relevant experience in software/ERP development for Public Sector Organizations	State owned company/organization incorporation date and verifiable project history spanning at least 5 years.	Pass / Fail
E-03	Not blacklisted by any federal or provincial government agency in Pakistan.	Self-declaration on company/organization letterhead, signed by an authorized representative.	Pass / Fail
E-04	Technical + Financial Proposal	Submission of complete proposal (Technical + Financial separately)	Pass / Fail
E-05	Compliance with RFP Instructions	Technical and Financial proposal evaluation	Pass / Fail

8.2 Joint Ventures and Sub-Contracting

Joint ventures and sub-contracting are not permitted. The bidder must be a single legal entity that independently meets all eligibility criteria in Section 8.1 and shall execute all work using its own employees. FPSC requires a single entity bearing full responsibility for quality, timelines, and deliverables throughout the project lifecycle and warranty period.

9. Technical Evaluation Criteria

Only bidders who pass all mandatory eligibility criteria in Section 8 will proceed to technical evaluation. The evaluation committee, comprising FPSC and PISC representatives, will score each technically eligible proposal against the criteria below.

Minimum qualifying technical score: 70 out of 100. Only bidders scoring 70 or above will have their financial proposals opened.

9.1 Scoring Matrix

Ref	Sub-Criterion	Scoring Guidance	Max	Evidence
T-01	Relevant experience in ERP development and operationalization for Public Sector Organizations	Experience of ERP development and operationalization - up to 5 years: 10 marks - up to 10 years: 12 marks - more than 10 years: 15 marks	15	Completion certificates or client attestation letters.
T-02	Deployed QDB & CBT system capability	Scored on the T-02 detailed rubric in Section 9.1.1: number of live deployments, scale of deployment, recency of active use, and CBT/QDB domain relevance.	20	Completion certificates or client attestation letters.
T-03	Recruitment cycle automation achieved	Scored on the T-03 detailed rubric in Section 9.1.1: extent of recruitment cycle automation, scale of public-sector usage, volume and operational load, and process intelligence.	15	Completion certificates or client attestation letters.
T-04	Key personnel and ERP-relevant team	Scored on the T-04 detailed rubric in Section 9.1.1: team completeness, key seniority sufficiency, and domain-specific experience alignment (see Section 9.2).	15	Appointment letters and CVs/certification of resources
T-05	Work Plan & Methodology	- Technical proposal	15	Technical Proposal
T-06	Demonstration / Presentation	Scored on the T-06 detailed rubric in Section 9.1.1: end-to-end system demonstration, usability and workflow efficiency, and scalability and performance evidence.	10	Presentation
T-07	Innovation and government capacity building	Scored on the T-07 detailed rubric in Section 9.1.1: AI-based automation capability; and government capacity building and training capability.	10	Technical documentation, system demonstration, training framework, and government client references.

9.1.1 Detailed Scoring Rubrics

Criteria T-02, T-03, T-04, T-06, and T-07 are scored on the detailed rubrics below. Marks are awarded only on demonstrated, verifiable evidence; the evaluation committee may award intermediate marks within a band where a bidder partly meets the next level. Criteria T-01 and T-05 are scored as set out in the matrix above.

T-02 Deployed QDB & CBT System Capability (20 marks)

This criterion scores the bidder's Question Data Bank and Computer-Based Testing systems that are already developed, deployed, and in live operation.

Scoring Band	Descriptor	Marks
T-02.1 Number of Live Deployments (maximum 8 marks) — count only QDB/CBT systems in live operational use; pilots are excluded unless live use is confirmed.		
1 to 2 systems	One to two QDB/CBT systems in live operational use	5
3 to 5 systems	Three to five systems in live operational use	7
More than 5	More than five systems in live operational use	8
T-02.2 Scale of Deployment, Pakistan context (maximum 5 marks)		
Small	Up to 1,000 users per year, or a single institution	2
Medium	1,000 to 25,000 users per year, provincial scale	3
Large	More than 25,000 users per year, federal or multi-provincial scale	5
T-02.3 Recency of Active Use (maximum 4 marks)		
Inactive >5 years	Last known active use more than 5 years ago	1
2 to 5 years	Last active use 2 to 5 years ago	2
Within 2 years	Active use within the last 2 years	4
T-02.4 Domain Relevance, CBT/QDB fit (maximum 3 marks)		
Low	Generic quizzes or surveys	1
Moderate	Academic examinations with basic CBT features	2
High	Secure high-stakes CBT with question-bank governance	3

T-03 Recruitment Cycle Automation Achieved (15 marks)

This criterion scores the level of automation the bidder has practically achieved in recruitment and examination systems it has developed or owns.

Scoring Band	Descriptor	Marks
T-03.1 Extent of Recruitment Cycle Automation (maximum 6 marks)		
Low	Only partial modules automated, e.g. application collection only	2
Medium	Multiple stages automated: application, shortlisting, scheduling	4
High	End-to-end automation: application, screening, testing, result, merit-list generation	6
T-03.2 Scale of Public-Sector Usage (maximum 5 marks)		
Small	Local bodies or single-institution recruitment	2
Medium	Provincial departments, universities, or boards	3
Large	Federal or multi-provincial recruitment bodies	5
T-03.3 Volume and Operational Load (maximum 3 marks)		
Low	Fewer than 5,000 applicants per year	1
Medium	5,000 to 50,000 applicants per year	2
High	More than 50,000 applicants per year with concurrent processing	3
T-03.4 Process Intelligence and Automation Sophistication (maximum 1 mark)		
Basic	Rule-based workflows only	0
Advanced	Auto-merit, eligibility rules engine, anomaly detection, or decision automation	1

T-04 Key Personnel and ERP-Relevant Team (15 marks)

This criterion scores the proposed team for fitness to develop an ERP for FPSC. It assesses whether the team composition, seniority, and domain experience match what ERP, CBT, and recruitment system development actually requires (see Section 9.2 for the key positions).

Scoring Band	Descriptor	Marks
T-04.1 Team Completeness (maximum 5 marks)		
Low	Missing three or more critical roles, or a weak team structure	2
Medium	All core roles present, with some gaps in supporting roles	4
High	All required roles present: ERP, CBT, recruitment, infrastructure, QA, and security	5
T-04.2 Key Seniority Sufficiency (maximum 5 marks)		
Low	Experience below threshold in two or more key roles	2
Medium	Meets minimum experience in most key roles	3 to 4
High	Exceeds experience requirements in all key roles	5
T-04.3 Domain-Specific Experience Alignment (maximum 5 marks)		
Low	Generic IT or ERP experience only	2
Medium	Some public-sector or education-systems experience	3 to 4
High	Direct experience in ERP, CBT, and recruitment or government systems	5

T-06 Demonstration and Presentation (10 marks)

The demonstration is scored objectively against fixed, observable bands. Each bidder demonstrates a live system so that demonstrations are comparable on the same scale.

Scoring Band	Descriptor	Marks
T-06.1 End-to-End System Demonstration (maximum 5 marks) — evidence: live system demonstration.		
Low	Isolated modules or partial features only, e.g. login or forms	1 to 2
Medium	Partial workflow, e.g. application to CBT, or a single recruitment module	3 to 4
High	Fully integrated end-to-end workflow: ERP, QDB, CBT, and recruitment cycle	5
T-06.2 Usability and Workflow Efficiency (maximum 3 marks)		
Low	Complex navigation, multiple manual steps, non-intuitive interface	1
Medium	Moderate usability with some manual steps	2
High	Streamlined workflows, role-based dashboards, minimal clicks	3
T-06.3 Scalability and Performance Evidence (maximum 2 marks)		
Low	No evidence of scalability, or weak performance during the demonstration	0
Medium	Demonstrates moderate or simulated load handling	1
High	Demonstrates high-volume concurrent usage or large-scale simulation	2

T-07 Innovation and Government Capacity Building (10 marks)

This criterion scores two factors: the maturity of AI-based automation in the bidder's systems, and the bidder's proven ability to build capacity in government users.

Scoring Band	Descriptor	Marks
T-07.1 AI-Based Automation Capability (maximum 5 marks) — evidence: technical documentation, system demonstration, project reports.		
Low	No demonstrated AI/ML in automation; traditional rule-based systems only	1 to 2
Medium	Limited AI use, e.g. basic analytics or simple recommendation features	3 to 4
High	Advanced AI integration, e.g. intelligent decision support, adaptive testing, anomaly detection	5
T-07.2 Government Capacity Building and Training Capability (maximum 5 marks) — evidence: training framework, plans, attendance records, client references.		
Low	No structured training programs; ad-hoc user training only	1 to 2
Medium	Structured training programs with documented sessions and materials	3 to 4

High

Comprehensive capacity-building framework: training academies, structured certification, client references

5

9.2 Key Experts / Staffing Requirements

The bidder shall propose a qualified and experienced team of key experts for successful execution of the project. The proposed team shall remain committed throughout the project duration and any replacement shall require prior approval of FPSC. The team include:

Sr. #	Position	Key Responsibilities	Minimum Qualification	Minimum Experience
i.	Project Manager / Team Lead	Overall project management, stakeholder coordination, delivery timelines	Bachelor's/Master's in IT/CS and relevant valid certification	10+ years
ii.	Solution / System Architect	Design system architecture, scalability, integration	Bachelor's/Master's in IT/CS and relevant valid certification	8+ years
iii.	Business Analyst(s)	Requirement gathering, BPR, documentation	Bachelor's in IT/Business and relevant valid certification	5+ years
iv.	Software Developers (Frontend/Backend)	Development of ERP, QDB, CBT modules	Bachelor's in IT/CS and relevant valid certification	3–5+ years
v.	Database Administrator (DBA)	Database design, optimization, backup, migration	Bachelor's in IT/CS and relevant valid certification	5+ years
vi.	Cyber Security Expert	Security implementation, audits, risk assessment	Bachelor's in IT/CS and relevant valid certification	5+ years
vii.	QA / Testing Engineers	System testing, UAT coordination, defect tracking	Bachelor's in IT/CS and relevant valid certification	3–5+ years
viii.	UI/UX Designer	Interface design, usability, accessibility	Bachelor's in Design/IT and relevant valid certification	3+ years
ix.	DevOps / System Engineer	Deployment, CI/CD, infrastructure management	Bachelor's in IT/CS and relevant valid certification	5+ years
x.	Training & Change	User training,	Bachelor's in	5+ years

Sr. #	Position	Key Responsibilities	Minimum Qualification	Minimum Experience
	Management Specialist	documentation, system adoption	relevant field	

The bidder may propose additional resources as required to ensure timely completion and quality delivery.

9.3 Evaluation Process

1. The evaluation committee opens Technical Proposals (Envelope 1) on the scheduled date in the presence of bidders or their representatives who choose to attend.
2. The committee verifies mandatory eligibility (Section 8). Bidders failing any criterion are disqualified and their Financial Proposals are returned unopened.
3. Eligible bidders' proposals are scored against the proposal- and evidence-based criteria T-01 to T-05 and T-07 using the scoring matrix and the detailed rubrics in Section 9.1. Criterion T-06 (Demonstration and Presentation) is scored separately at the demonstration stage.
4. Bidders shortlisted after proposal scoring shall be invited for Demonstration and Presentation (criterion 6). At least 7 working days' notice shall be provided. Each bidder is allocated a minimum of 60 minutes (30 minutes demo, 15 minutes presentation, 15 minutes Q and A).
5. The final technical score is the sum of the proposal-based scores (criteria T-01 to T-05 and T-07) and the demonstration score (criterion T-06), averaged across all committee members.
6. Bidders scoring below 70/100 overall are eliminated. Their Financial Proposals are returned unopened.
7. Financial Proposals of technically qualified bidders will be opened on a later stage. Date/time of financial bids opening shall be announced accordingly.
8. FPSC reserves the right to seek written clarifications and to verify all claims, including contacting client references and requesting certification evidence.

9.4 Scoring Notes

Marks are awarded based on demonstrated evidence, not claims. Unsubstantiated assertions receive zero marks. The evaluation committee may award intermediate scores within the stated ranges based on quality and completeness of evidence.

10. Proposal Submission

10.1 Submission Method

Proposals shall be submitted using the single-stage, two-envelope method in accordance with PPRA Rule 36(b). Each bidder shall submit two separately sealed envelopes, clearly marked, placed together in one outer sealed envelope.

Item	Details
Outer Envelope	Sealed, marked: "RFP: Software Development Services, FPSC/PISC/SW/2026/01". Contains Envelope 1 and Envelope 2. Bidder name and address on the back.
Envelope 1	Sealed, marked: "TECHNICAL PROPOSAL, DO NOT OPEN WITH FINANCIAL PROPOSAL". Contains the technical proposal as specified in Section 10.2.
Envelope 2	Sealed, marked: "FINANCIAL PROPOSAL". Contains the financial proposal as specified in Section 10.3.

10.2 Technical Proposal Contents (Envelope 1)

The technical proposal shall contain the following, organized in the order listed:

- Company/organization profile: organizational structure, ownership, office locations, total staff count, software development team size, and brief company/organization history.
- Eligibility evidence: all documents specified in Section 8.1 (E-01 through E-05) in the order listed. Missing eligibility documents result in disqualification.
- Technical approach: detailed description of the proposed solution including system architecture (aligned to the 4-tier model in Section 4), technology stack with justification, security design, QDB protection approach, offline CBT/edge node design, scalability strategy, and data migration approach.
- Compliance matrix: completed Annexure A responding to every functional requirement in Section 3, every non-functional requirement in Section 5, and every technical requirement in Section 6, with: Fully Compliant / Partially Compliant (with explanation) / Non-Compliant.
- Key personnel: CVs of the key positions specified in Section 9.2 using the format in Annexure C. Attach copies of certifications claimed.
- Work plan: Gantt chart showing task breakdown, dependencies, parallel workstreams, resource allocation by name, and milestones aligned to Section 7.1 phases.
- Risk register: identified risks specific to this project with probability, impact assessment, mitigation strategy, and contingency plan.
- QA methodology: testing strategy covering unit, integration, system, security, performance, and UAT testing. Defect management process. Code review approach.
- Client references: at least 2 references for similar projects. Each reference must include: project name, client organization, project value, duration, number of users

served, scope summary, and client contact person with phone and email for verification.

The technical proposal shall be submitted in 1 original (clearly marked "ORIGINAL") and **2 (two)** copies (marked "COPY"), plus 1 soft copy on USB drive in PDF format. In case of discrepancy between the original and copies, the original shall prevail.

10.3 Financial Proposal Contents (Envelope 2)

The financial proposal shall be submitted using the format provided in Annexure B and shall include:

- Total bid price in Pakistani Rupees (PKR), inclusive of all applicable taxes.
- Cost breakdown by module: separate line items for Module 1 (GR), Module 2 (CE), Module 3 (UEM), Module 4 (Website and Portal), Module 5 (CBT and QDB), and Module 6 (Supporting). Cross-cutting components (audit trail, notifications, reporting, APIs, workflow engine) shall be allocated proportionally across modules or shown as a separate line.
- Cost breakdown by phase: itemized costs for each phase in Section 7.1 (Inception/BPR/Design, Core EMS, CBT/QDBMS, Portal/Integration, Pilot, Full Deployment, Go-Live, Warranty).
- License costs: if any licensed software is proposed, itemize each license with: product name, license type (perpetual/subscription), unit cost, quantity, duration, and renewal terms.
- Third-party costs: any costs for third-party services or components (e.g., SMS gateway, payment gateway integration, SEB licenses) listed separately.
- Training costs: itemized per audience group as defined in Section 7.3.
- Warranty and O and M costs: 24-month post-go-live warranty support as a separate line item.
- Payment schedule: proposed milestone-based payment schedule linked to the phases in Section 7.1 and conditional on Milestone Acceptance Certificates (Section 7.4).

The financial proposal shall be submitted in 1 original (marked "ORIGINAL") and 1 copy (marked "COPY"), plus 1 soft copy on USB drive in PDF format.

Conditional pricing, price escalation clauses, or prices denominated in foreign currency are not acceptable. All prices shall be fixed for the duration of the contract.

10.4 Bid Security

Each bidder shall furnish, along with the Technical Proposal, a bid security equal to 3% (three percent) and performance security equal to (10%) of the total bid price quoted in the Financial Proposal in the form of CIDR/bank guarantee for 2yrs.

The bid security and performance security shall be furnished in the form of a Demand Draft, Pay Order, or an unconditional Bank Guarantee issued by a scheduled bank operating in Pakistan, drawn in favour of the Federal Public Service Commission. Cheques, cash, and conditional instruments are not acceptable.

The bid security shall remain valid for 90 days beyond the proposal validity period specified in Section 10.5. It shall be enclosed in a separate sealed envelope placed inside the Technical Proposal envelope (Envelope 1). A proposal not accompanied by the required bid security, in the prescribed form and amount, shall be rejected as non-responsive without further evaluation. Whereas performance security is valid for the duration of the project.

The bid security of unsuccessful bidders shall be released after the contract is awarded to the successful bidder. The bid security of the successful bidder shall be released upon submission of the performance security and signing of the contract.

The bid security shall be forfeited if the bidder withdraws or modifies its proposal during the proposal validity period, or if the successful bidder fails to sign the contract or to furnish the performance security within the time stipulated in the Letter of Intent.

10.5 Submission Details

Item	Details
Submission Deadline	[Bids must be reached by 02:00 PM on 08-June-2026 and opened at 02:30 PM same day]. Pakistan Standard Time (PST). Proposals received after this time will not be accepted under any circumstances, in accordance with PPRA Rules.
Submission Address	[FPSC, Aga Khan Road, Sector F-5/1, Islamabad]
Addressed To	[Project Director, FPSC]
Proposal Validity	Proposals shall remain valid for [60] days from the submission deadline. FPSC may request an extension of validity, which the bidder may accept or reject.

10.6 Pre-Bid Conference

Item	Details
Date and Time	[03-June-2026], [02:30] PST.
Venue	FPSC Headquarters, Aga Khan Road, F-5/1, Islamabad.
Purpose	Clarification of RFP requirements. Walkthrough of deployment architecture. Q and A session.
Attendance	Recommended but not mandatory. Non-attendance does not disqualify a bidder.

10.7 Clarifications and Amendments

Prospective bidders may submit written questions or requests for clarification regarding this RFP. The process is as follows:

- Questions shall be submitted in writing via email to it@fpssc.gov.pk no later than **3rd June 2026**.
- FPSC will compile all questions received (without identifying the questioner) and issue a consolidated written response to all prospective bidders who have obtained the RFP document.
- The consolidated response shall be issued on same day.
- Only written clarifications issued by FPSC are binding. Oral statements made during the pre-bid conference or at any other time do not modify this RFP.
- FPSC reserves the right to amend this RFP at any time before the submission deadline. Amendments will be communicated in writing to all prospective bidders and the submission deadline may be extended accordingly.

11. Evaluation Methodology

11.1 Method

Evaluation shall follow the Quality and Cost-Based Selection (QCBS) method as permitted under PPRA Rules. This method gives predominant weight to technical quality while considering cost competitiveness.

11.2 Weightages

The technical and financial proposals shall be weighted 70% and 30% respectively. The predominant weight on technical quality reflects the sensitivity and national importance of the FPSC examination and recruitment platform, where capability, security, and reliability outweigh marginal price differences. Technical Weight = 0.70; Financial Weight = 0.30.

Component	Weight
Technical Proposal	70%
Financial Proposal	30%

11.3 Technical Score

The technical score for each qualified bidder is determined through the evaluation process described in Section 9. Only bidders achieving the minimum qualifying score of 70 out of 100 proceed to financial evaluation.

11.4 Financial Score

The financial score is calculated using the following formula:

$$\text{Financial Score} = (\text{Lowest Bid Price among qualified bidders} / \text{Bidder's Bid Price}) \times 100$$

Where the Lowest Bid Price is the lowest total bid price among all technically qualified bidders. The bidder with the lowest price receives a financial score of 100. All other bidders receive proportionally lower financial scores.

11.5 Combined Score

The final combined score is calculated as:

$$\text{Combined Score} = (\text{Technical Score} \times \text{Technical Weight}) + (\text{Financial Score} \times \text{Financial Weight})$$

11.6 Illustrative Example

The following example illustrates the QCBS calculation assuming a 70:30 technical-to-financial weightage:

Bidder	Technical Score (/100)	Bid Price (PKR M)	Financial Score	Combined Score	Rank
A	85	180	$(150/180) \times 100 = 83.3$	$(85 \times 0.70) + (83.3 \times 0.30) = 84.49$	2nd

Bidder	Technical Score (/100)	Bid Price (PKR M)	Financial Score	Combined Score	Rank
B	90	220	$(150/220) \times 100 = 68.2$	$(90 \times 0.70) + (68.2 \times 0.30) = 83.46$	1st
C	75	150	$(150/150) \times 100 = 100.0$	$(75 \times 0.70) + (100 \times 0.30) = 82.50$	3rd

In this example, Bidder B wins. Under an 70:30 weighting, technical quality is decisive: although Bidder C offers the lowest price and the maximum financial score, Bidder B's superior technical score produces the highest combined score.

11.7 Award

The bidder with the highest combined score shall be recommended for contract award, subject to the following:

- The recommendation is made by the evaluation committee.
- Upon approval by PAO, FPSC issues a Letter of Intent (LOI) to the successful bidder.
- Contract negotiations, if required, shall not alter the fundamental scope, terms, or price of the proposal.
- If contract negotiations fail, FPSC may invite the next-ranked bidder for negotiations.
- FPSC reserves the right to reject all proposals if none meet the requirements, in accordance with PPRA Rules.
- All bidders shall be notified of the outcome in writing. Unsuccessful bidders may request a debriefing within 15 days of notification.

12. Key Commercial Terms

The full contractual terms are specified in the Draft Contract Template (Annexure D). Submission of a proposal constitutes acceptance of these terms in principle. The following key terms directly affect proposal pricing and the Vendor must account for them:

(a) Contract Duration: 12 months for software development from contract signing, followed by a 24-month warranty and operations support period commencing from the date of Go-Live Acceptance Certificate. The warranty period is distinct from the development contract period.

(b) Payment: Milestone-based payments, conditional on issuance of Milestone Acceptance Certificates per Section 7.4. No advance payment. Payment released only after PISC review and FPSC acceptance of each deliverable.

(c) Liquidated Damages: To ensure timely completion of the Project deliverables, the Contractor/Vendor shall be liable to pay a penalty, on a monthly basis, for any delay in completion of the relevant project component beyond the stipulated Deliverable Completion Date, in accordance with the following formula:

$$\text{Monthly Penalty} = (\text{Cost of the Related Deliverable}) \times 0.09\%$$

(d) Intellectual Property: All source code, documentation, designs, databases, algorithms, configurations, and deliverables produced under this contract are the sole and exclusive property of FPSC. The Vendor shall execute an IP assignment agreement at contract signing. The Vendor shall not retain, reuse, or disclose any project output. Open-source components used shall be disclosed with license compatibility confirmation.

(e) Confidentiality: NDA at contract signing covering all FPSC data including candidate personal information, examination content, internal business processes, system designs, and source code. Question Data Bank content is subject to the highest level of confidentiality; any breach involving QDB data shall be treated as a material breach of contract. The NDA survives contract termination or completion.

(f) Warranty Obligations: During the 24-month warranty period, the Vendor shall fix all software defects at no additional cost, provide security patches and dependency updates quarterly (or immediately for critical vulnerabilities), participate in one annual DR drill at NTC Lahore, and facilitate one annual penetration test by an independent third party. Defect response and resolution service levels are specified in the Service Level Agreements in Section 7.5.

(g) Joint Ventures and Sub-Contracting: Not permitted. Requirements are specified in Section 8.2.

All other contractual provisions including personnel replacement, termination, force majeure, dispute resolution, and governing law are specified in Annexure D.

13. Annexures

The following annexures form part of this RFP. Templates marked with (*) shall be completed by the Vendor and submitted with the proposal.

Annexure	Title	Status
A *	Compliance Matrix Template: Vendor response to all functional (Section 3), non-functional (Section 5), and technical (Section 6) requirements using the revised requirement IDs.	Vendor completes and submits
B *	Financial Proposal Template: structured cost breakdown by module, by phase, licenses, training, and warranty.	Vendor completes and submits
C *	Key Personnel CV Template: standardized format for the key positions listed in Section 9.2, with fields for academic qualifications, relevant experience, certifications, and supporting evidence.	Vendor completes and submits
D	Draft Contract Template: includes all contractual terms covering personnel replacement, termination, force majeure, dispute resolution, and other standard provisions.	Provided by PISC (NUST), enclosed with this RFP

Annexure templates (A, B, C) will be provided in editable format (Excel/Word) along with this RFP document. Vendors must use the provided templates. Proposals submitted in non-standard formats may be rejected.

END OF RFP DOCUMENT